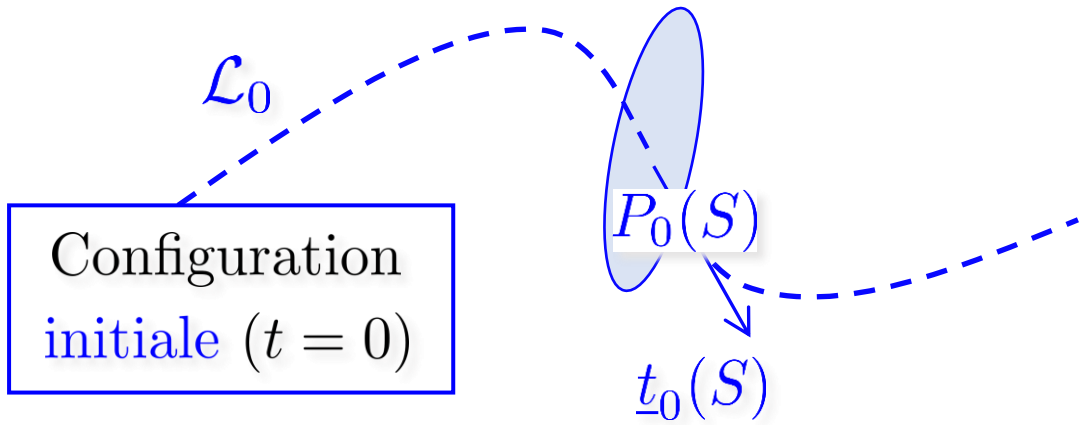
COURS DE MECANIQUE
Milieux curvilignes

Déformations, comportement élastique linéaire et méthode de résolution directe

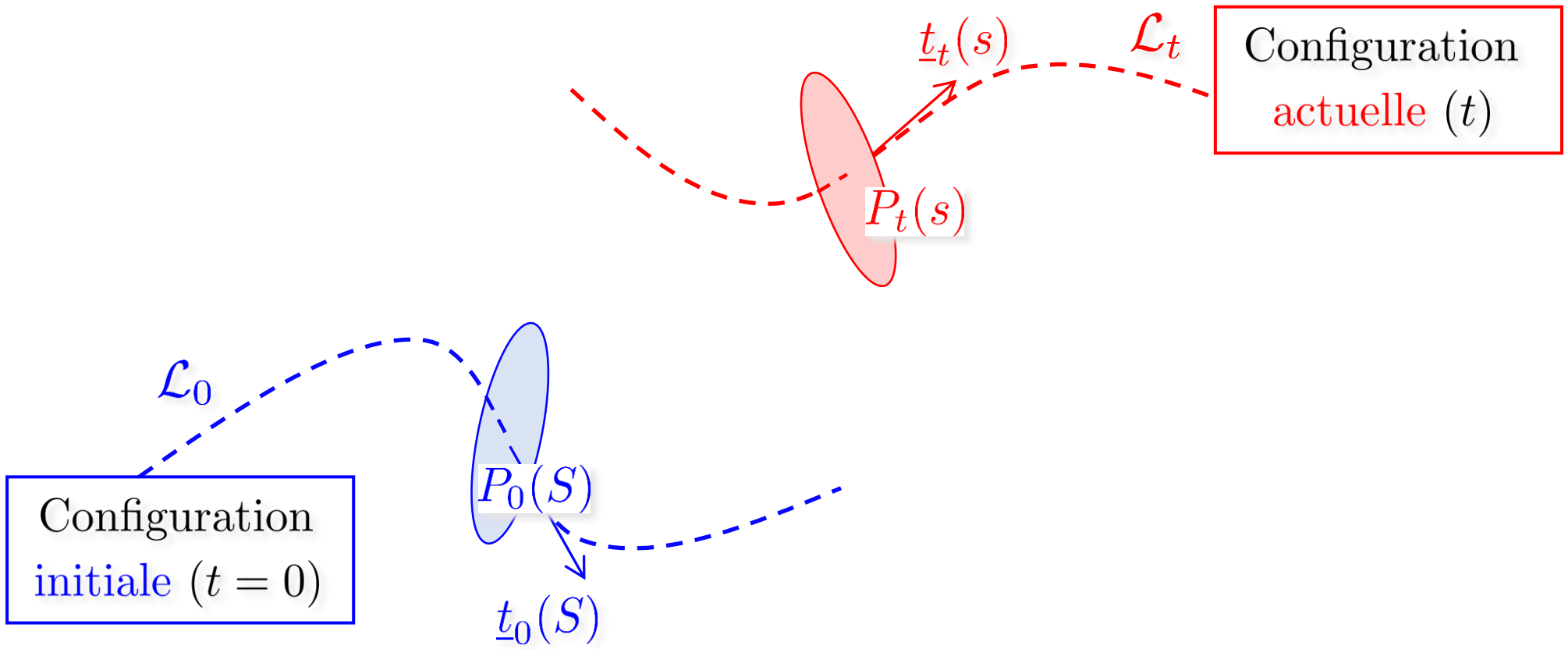
24 septembre 2026

Ecole Nationale des Ponts et Chaussées

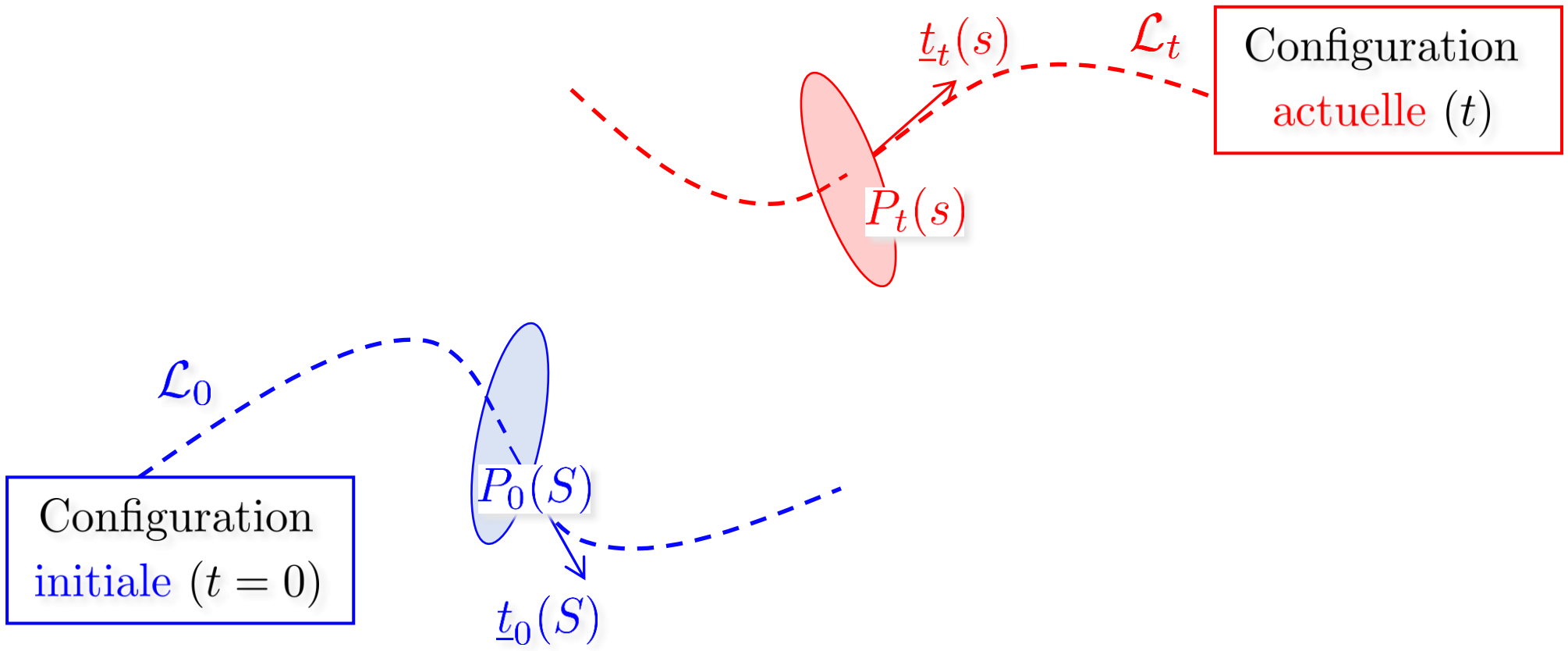
Transformation géométrique d'un arc (cinématique)



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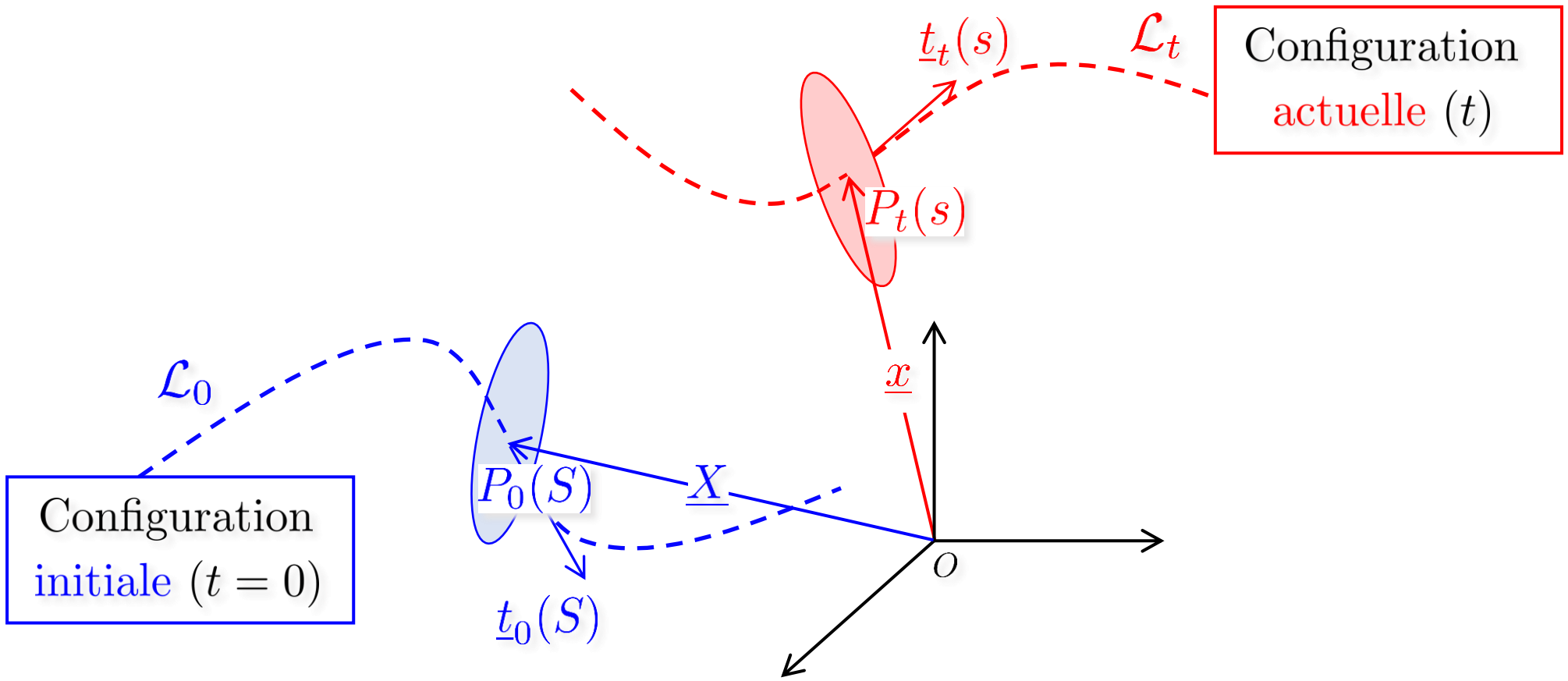


Transformation géométrique d'un arc (cinématique)



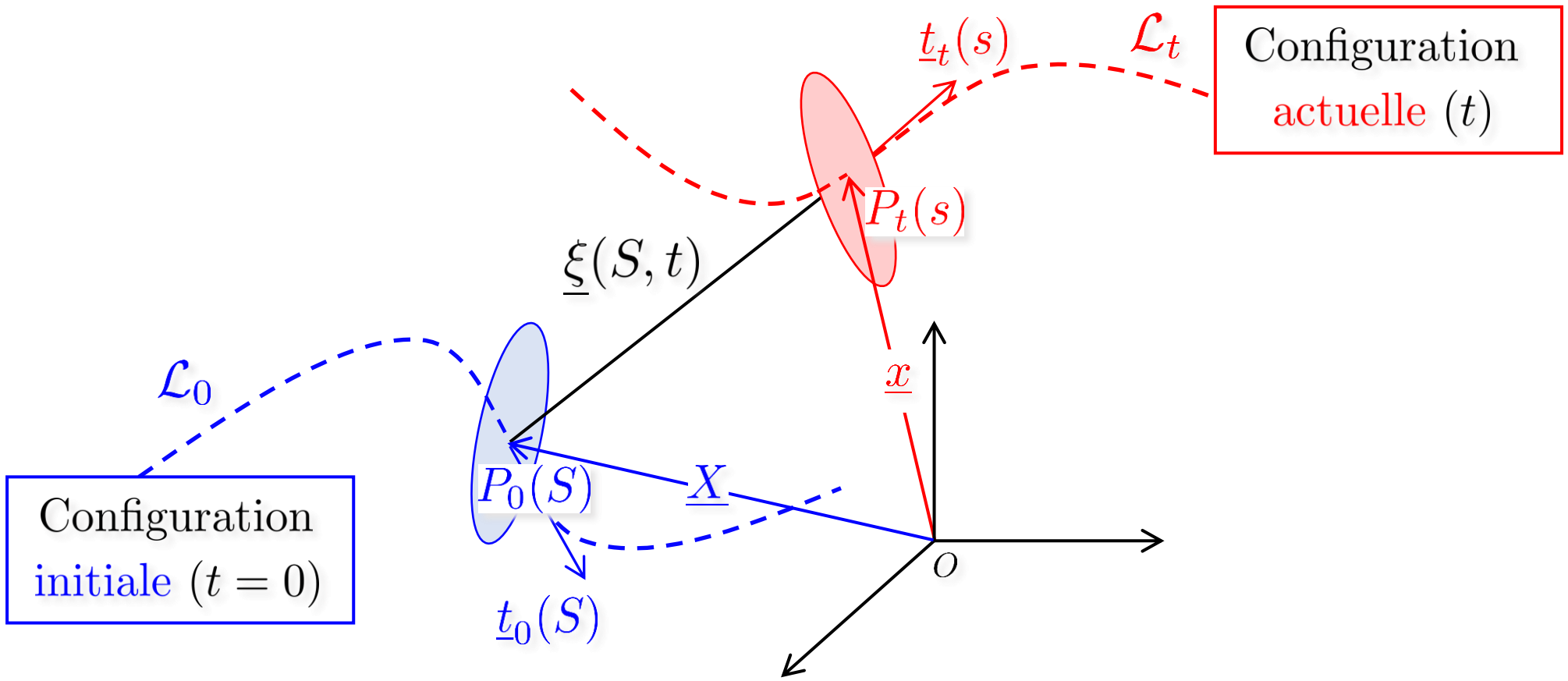
$\left.\begin{matrix} S \\ s \end{matrix}\right\}$ abscisse curviligne dans la $\begin{cases} \text{configuration initiale (de référence)} \\ \text{configuration actuelle (déformée)} \end{cases}$

Transformation géométrique d'un arc (cinématique)



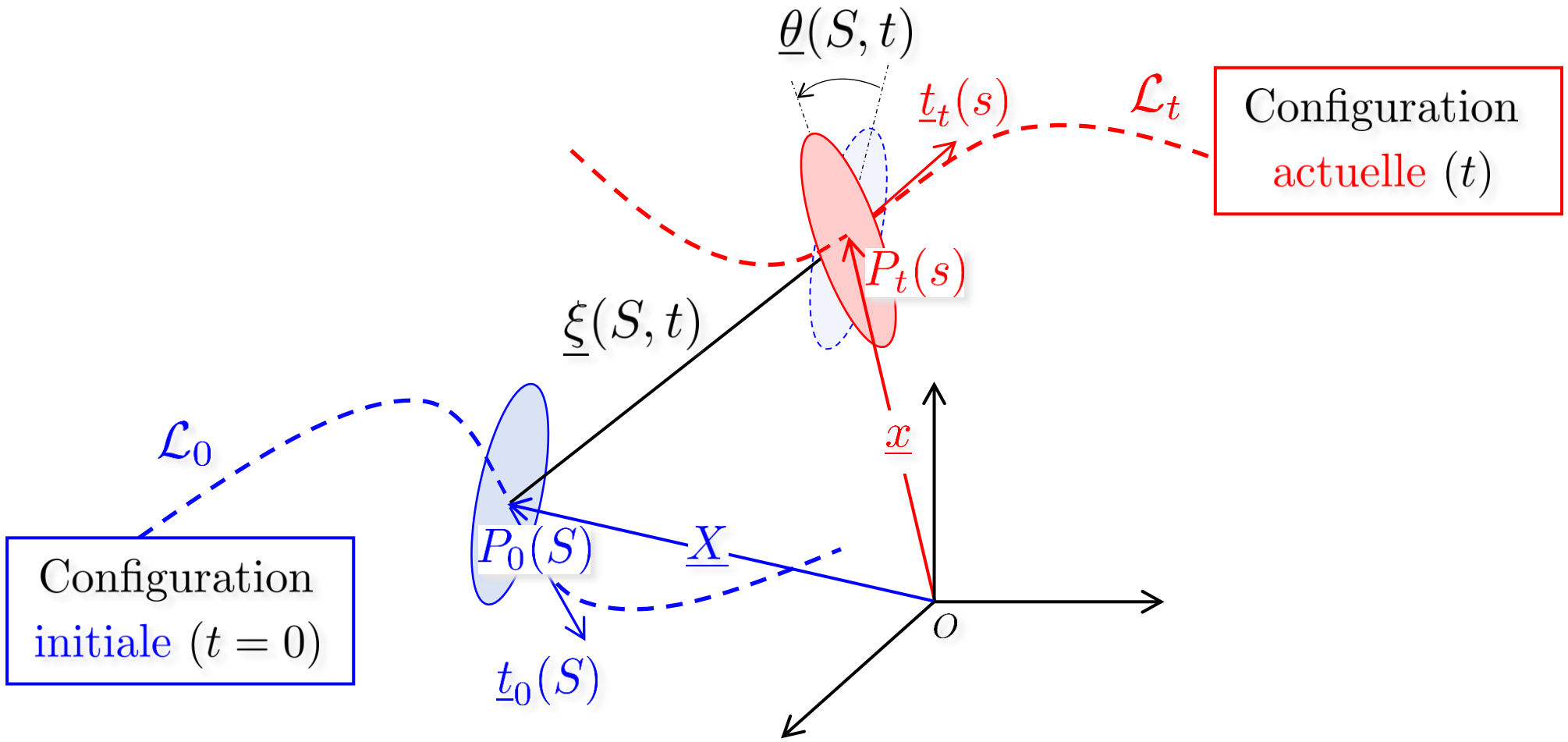
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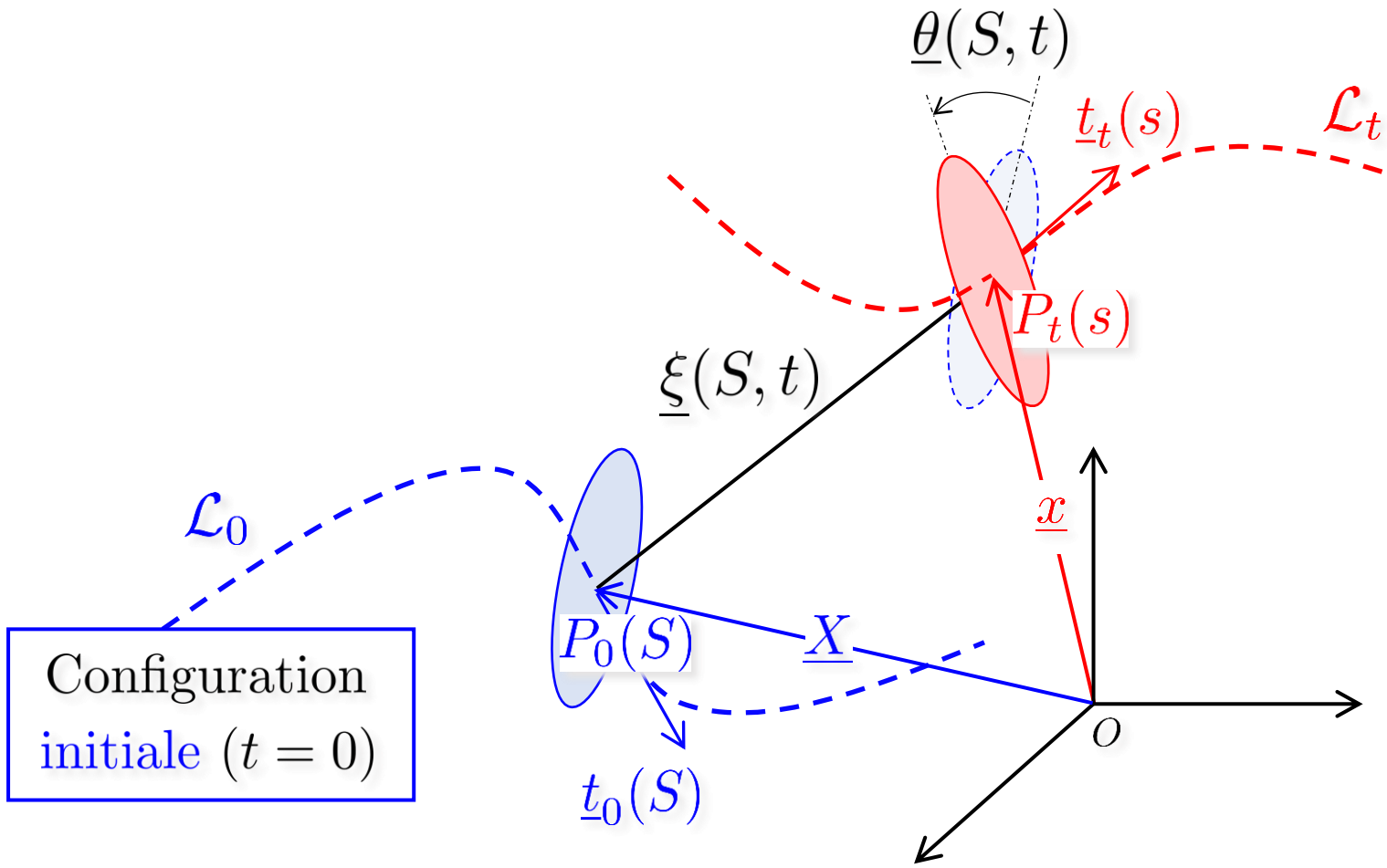
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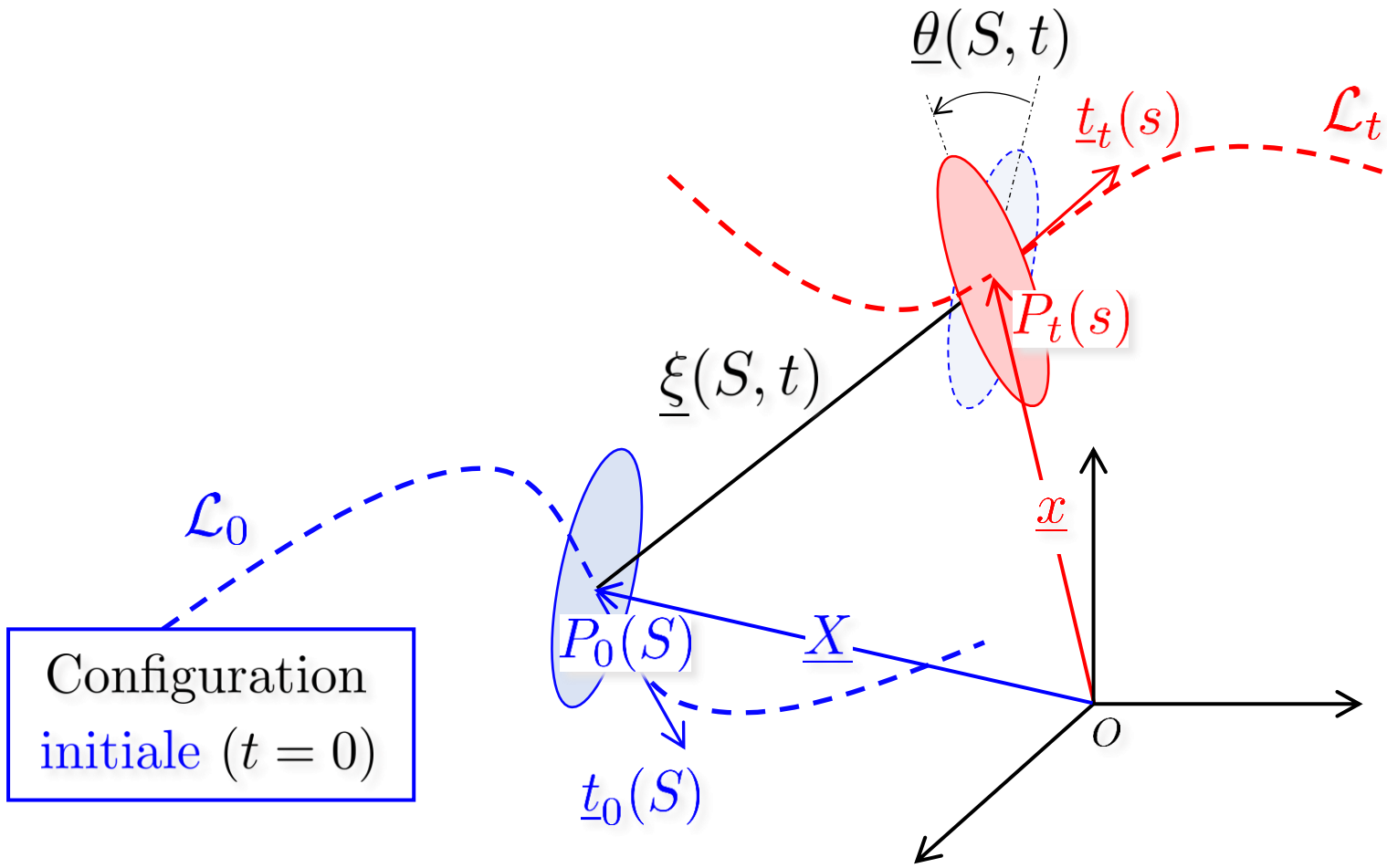
Transformation géométrique d'un arc (cinématique)



- $S \to \underline{\xi}(S,t)$: $\mathcal{C}_0$ et $\mathcal{C}_1^{pm}$
- $S \to \theta(S,t)$: $\mathcal{C}_0$ et $\mathcal{C}_1^{pm}$

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Transformation géométrique d'un arc (cinématique)



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- A $t=0$: $\begin{cases} s = S \\ \underline{\xi}(S,0) = 0 \\ \theta(S,0) = 0 \end{cases}$

$\left.\begin{matrix} S \\ s \end{matrix}\right\}$ abscisse curviligne dans la $\begin{cases} \text{configuration initiale (de référence)} \\ \text{configuration actuelle (déformée)} \end{cases}$

Cadre de travail : hypothèses et simplifications

Cadre de travail : hypothèses et simplifications

- structure plane

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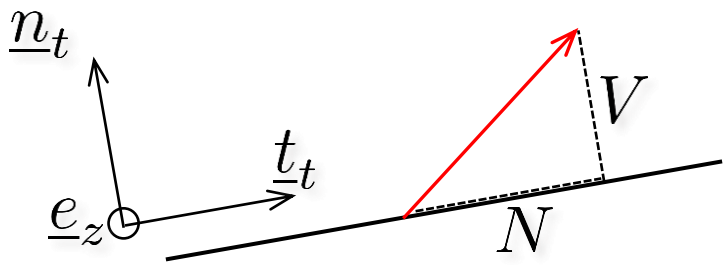
- structure plane
- tronçons rectilignes

Cadre de travail : hypothèses et simplifications

- structure plane
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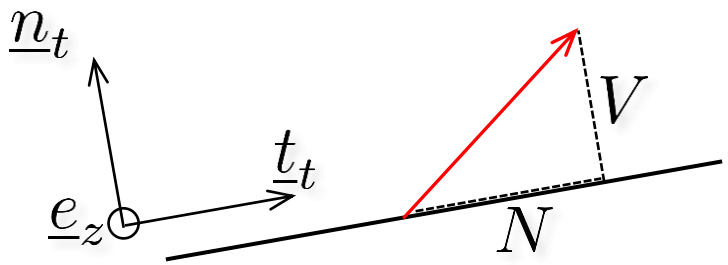
$$\underline{\Sigma}(s) = N(s)\underline{t}_t + V(s)\underline{n}_t \qquad \underline{\Gamma}(s) = M(s)\underline{e}_z$$

Cadre de travail : hypothèses et simplifications

- structure plane
- tronçons rectilignes
- chargement plan

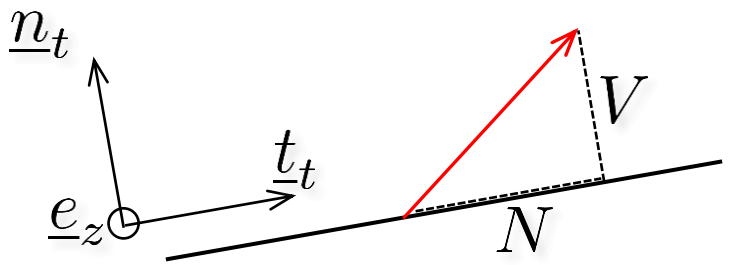
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- cinématique plane



Cadre de travail : hypothèses et simplifications

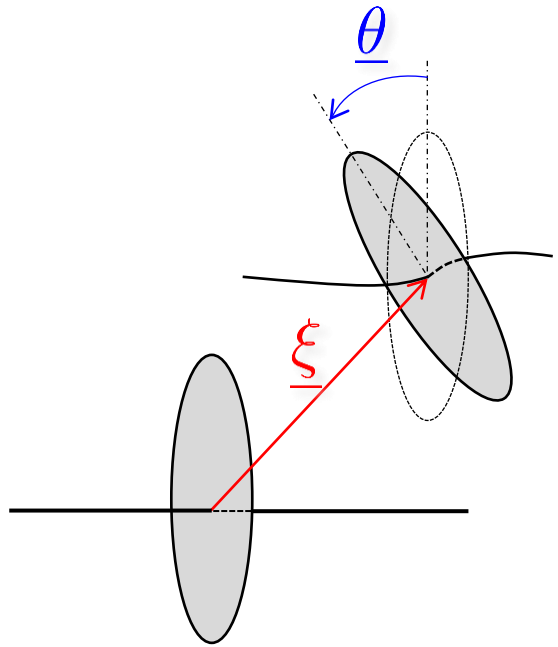
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$$\underline{\Sigma}(s) = N(s)\underline{t}_t + V(s)\underline{n}_t \qquad \underline{\Gamma}(s) = M(s)\underline{e}_z$$

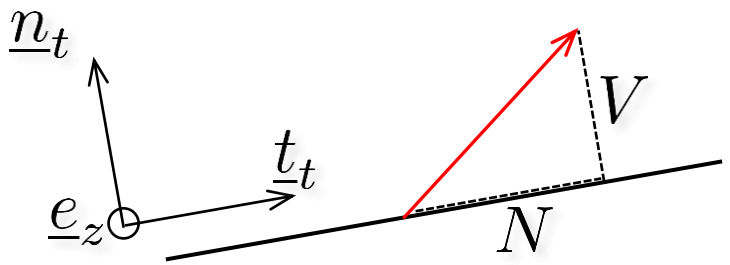
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$$\underline{\xi}(S) = \xi_t(S)\underline{t}_0 + \xi_n(S)\underline{n}_0$$
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Cadre de travail : hypothèses et simplifications

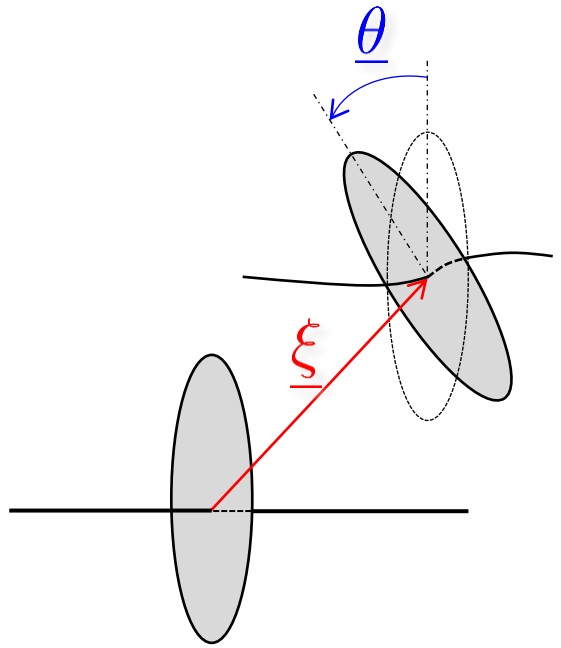
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$$\underline{\Sigma}(s) = N(s)\underline{t}_t + V(s)\underline{n}_t \qquad \underline{\Gamma}(s) = M(s)\underline{e}_z$$

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- transformation infinitésimale : $|\theta|, \left|\frac{\mathrm{d}\xi_t}{\mathrm{d}S}\right|, \left|\frac{\mathrm{d}\xi_n}{\mathrm{d}S}\right| \ll 1$

Déformation longitudinale

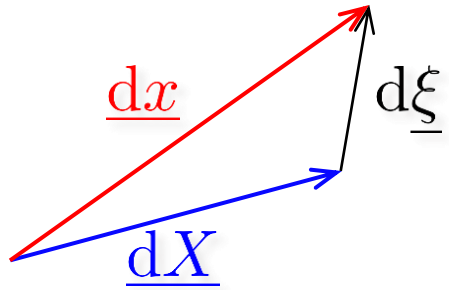
$$\underline{\mathrm{d}X} = |\underline{\mathrm{d}X}|\,\underline{t}_0 = \mathrm{d}S\;\underline{t}_0$$

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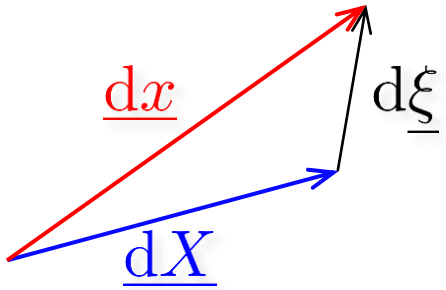
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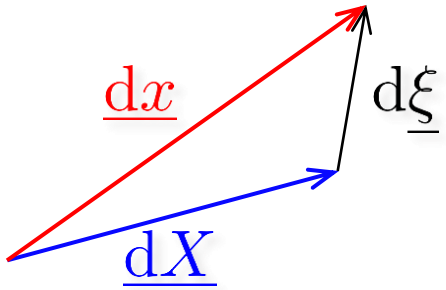


$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi}$$

Déformation longitudinale

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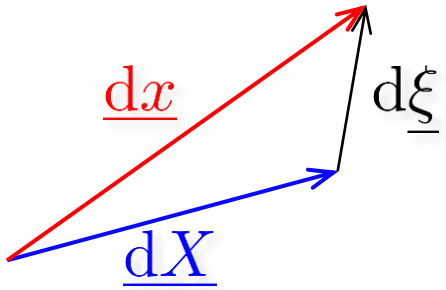


$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \quad \Longrightarrow \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

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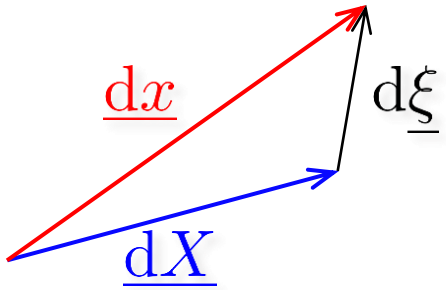
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dilatation (dans la direction $\underline{t}_0$) :

Déformation longitudinale

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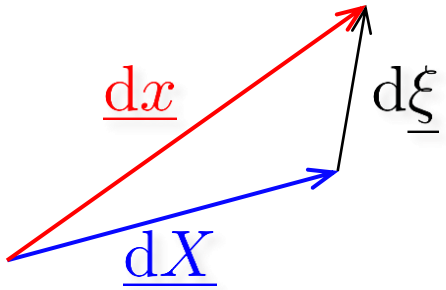
$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \quad \Longrightarrow \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

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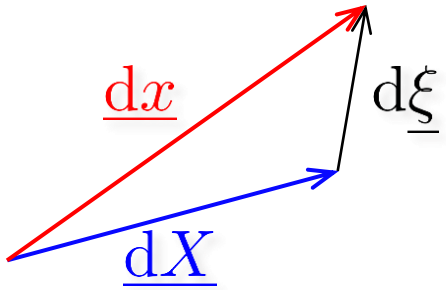
$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \quad \Longrightarrow \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

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Déformation longitudinale

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$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \quad \Longrightarrow \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

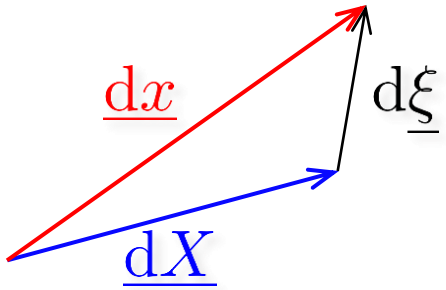
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déformation longitudinale (ou déformation d'extension): $\varepsilon = \dfrac{\mathrm{d}s - \mathrm{d}S}{\mathrm{d}S} = \dfrac{\mathrm{d}\xi_t}{\mathrm{d}S}$

Déformation longitudinale

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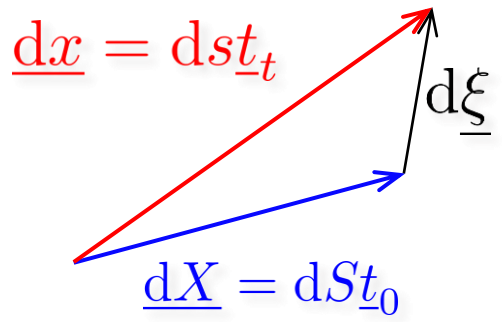
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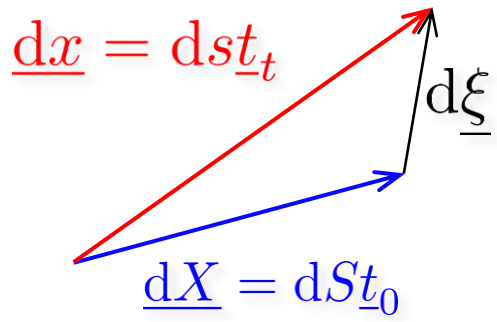
$$\frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S} = \frac{\mathrm{d}(\xi_t\underline{t}_0 + \xi_n\underline{n}_0)}{\mathrm{d}S} = \varepsilon\underline{t}_0 + \frac{\mathrm{d}\xi_n}{\mathrm{d}S}\underline{n}_0$$

Rotation de la tangente



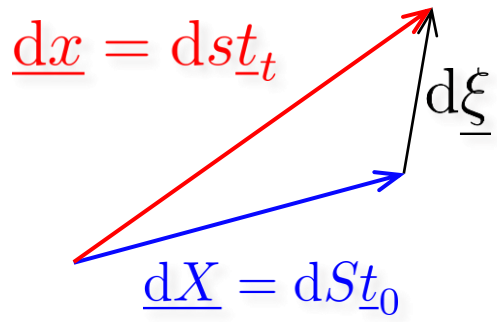
Rotation de la tangente

$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi}$$



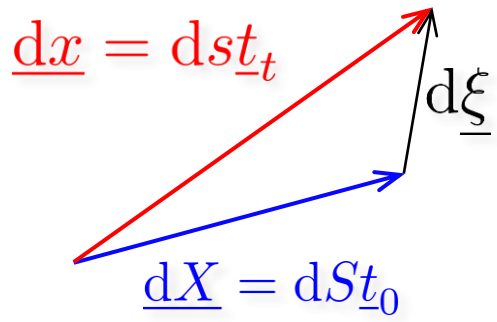
Rotation de la tangente

$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \implies \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t_0} + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$



Rotation de la tangente

$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \implies \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$



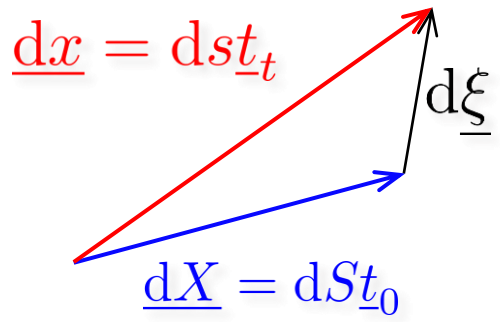
$$\implies \underline{t}_t = \frac{\mathrm{d}S}{\mathrm{d}s}\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

Rotation de la tangente

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$$\frac{1}{\lambda} = \frac{1}{1+\varepsilon} \to 1-\varepsilon$$

$$\implies \underline{t}_t = \boxed{\frac{\mathrm{d}S}{\mathrm{d}s}}\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

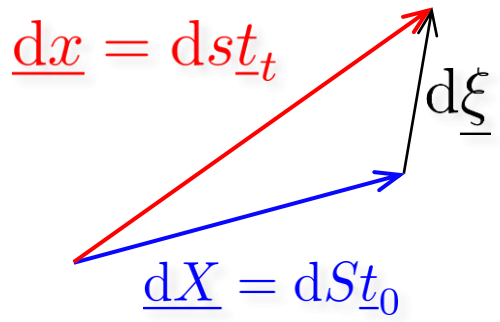


Rotation de la tangente

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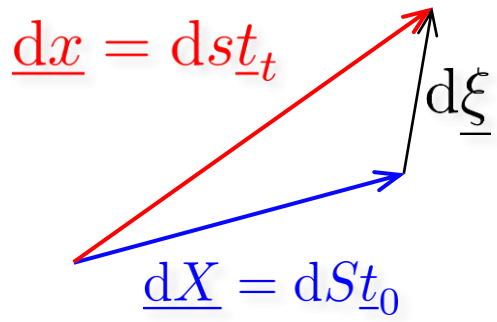


Rotation de la tangente

$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \implies \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

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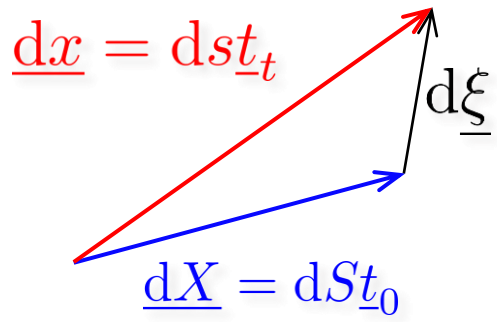


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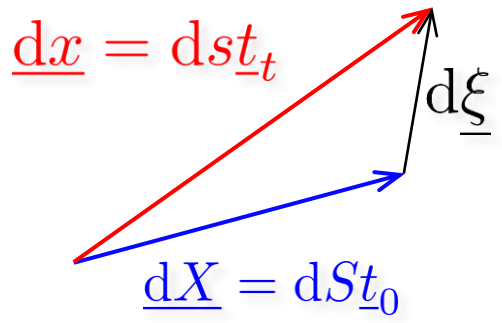


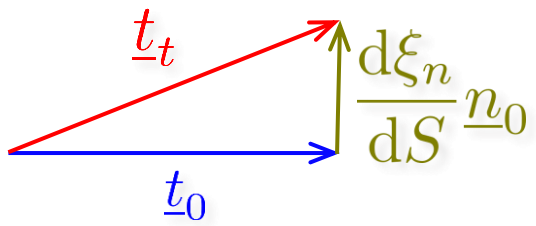
Rotation de la tangente

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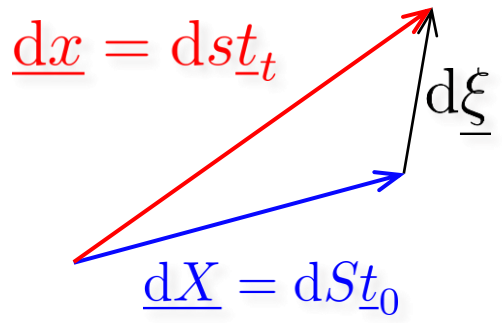


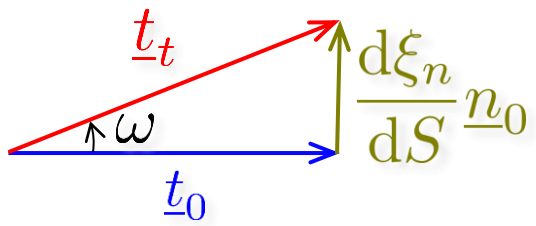
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$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \implies \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

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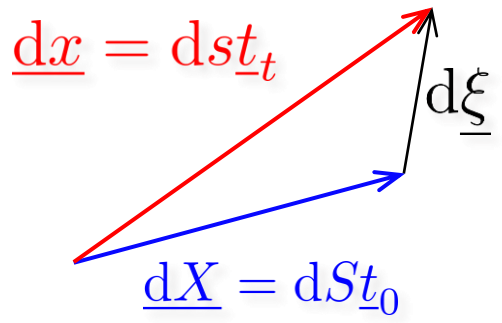
Rotation de la tangente

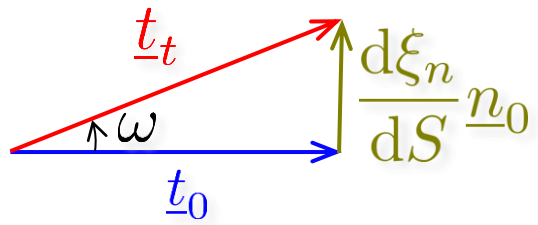
$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \implies \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

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$$\omega = \frac{\mathrm{d}\xi_n}{\mathrm{d}S}$$





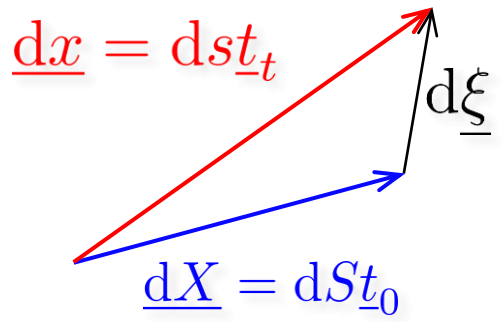
Rotation de la tangente

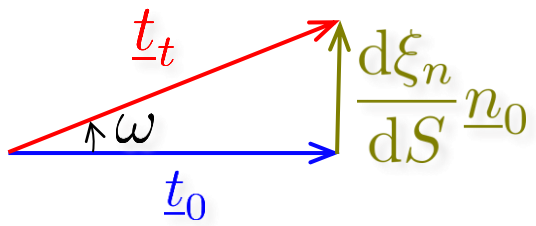
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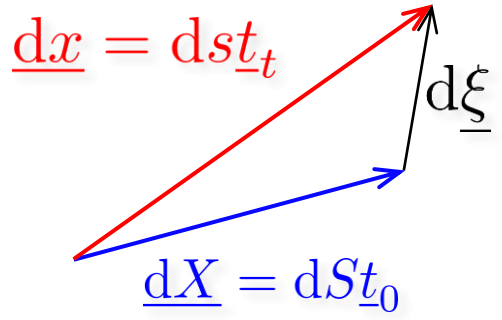
Rotation de la tangente

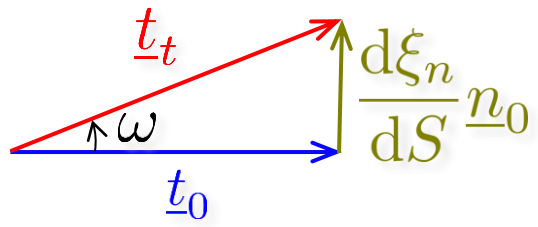
$$\underline{\mathrm{d}x} = \underline{\mathrm{d}X} + \mathrm{d}\underline{\xi} \implies \underline{\mathrm{d}x} = \mathrm{d}S\left(\underline{t}_0 + \frac{\mathrm{d}\underline{\xi}}{\mathrm{d}S}\right)$$

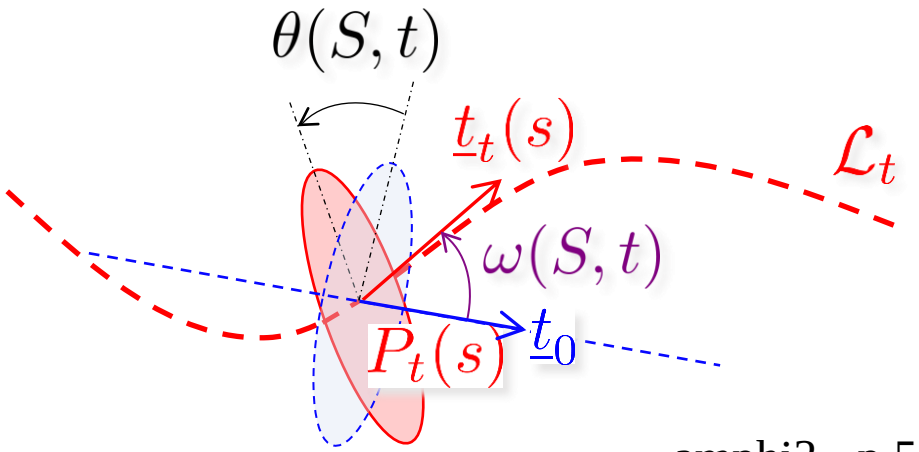
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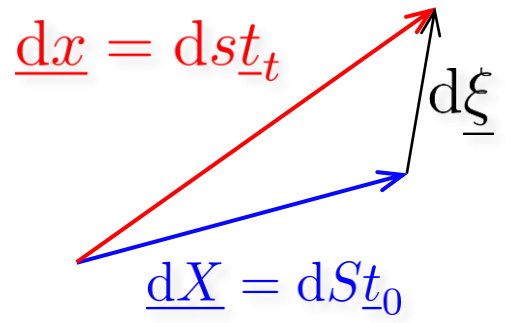


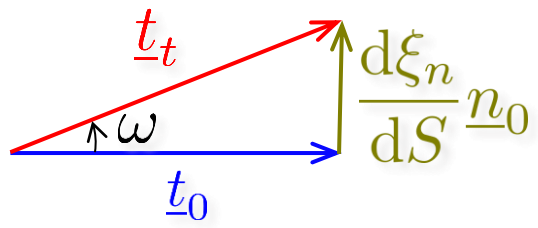
Rotation de la tangente

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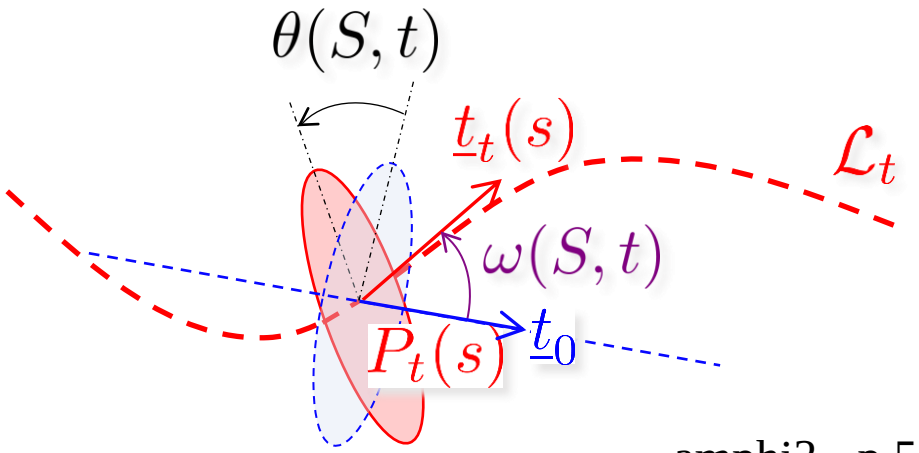
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$$\omega = \frac{\mathrm{d}\xi_n}{\mathrm{d}S} \qquad \underline{t}_t = \underline{t}_0 + \omega \underline{e}_z \wedge \underline{t}_0$$

Condition de Navier-Bernoulli : $\omega = \theta$

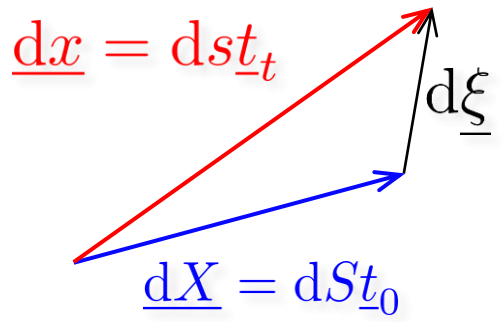


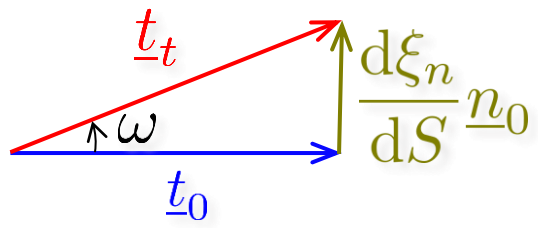
Rotation de la tangente

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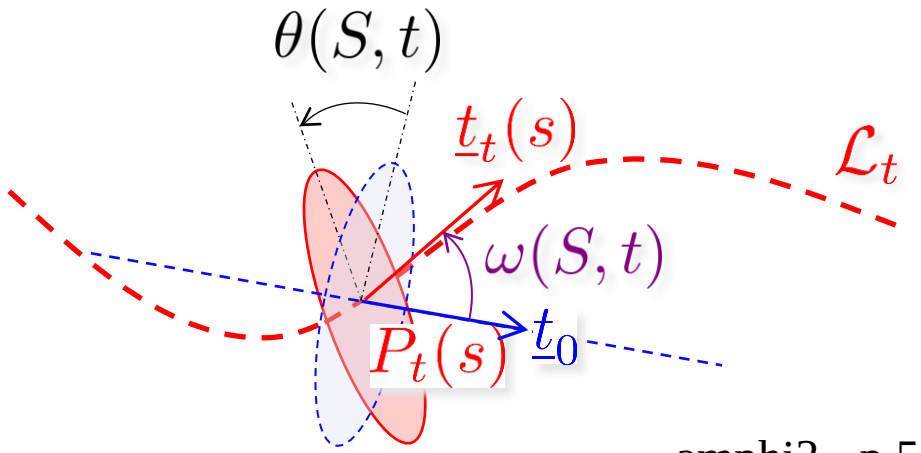




$$\omega = \frac{\mathrm{d}\xi_n}{\mathrm{d}S} \qquad \underline{t}_t = \underline{t}_0 + \omega \underline{e}_z \wedge \underline{t}_0$$

Condition de Navier-Bernoulli : $\omega = \theta$

déformation de cisaillement : $\omega - \theta = \frac{\mathrm{d}\xi_n}{\mathrm{d}S} - \theta$

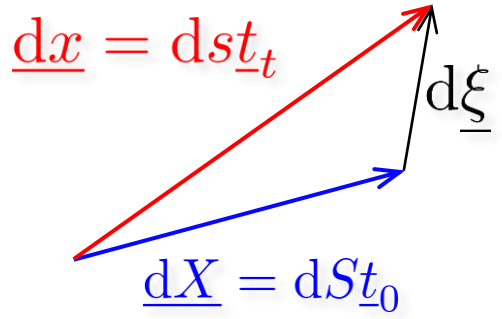


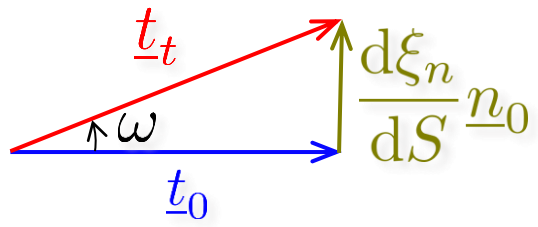
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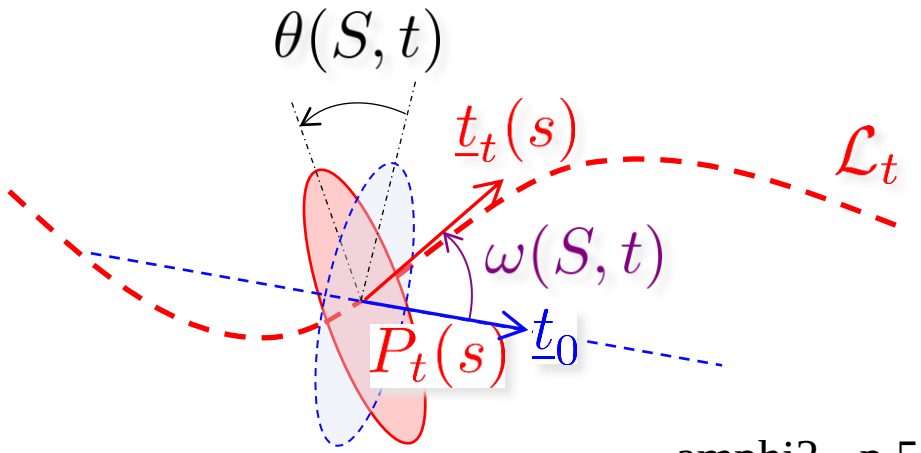




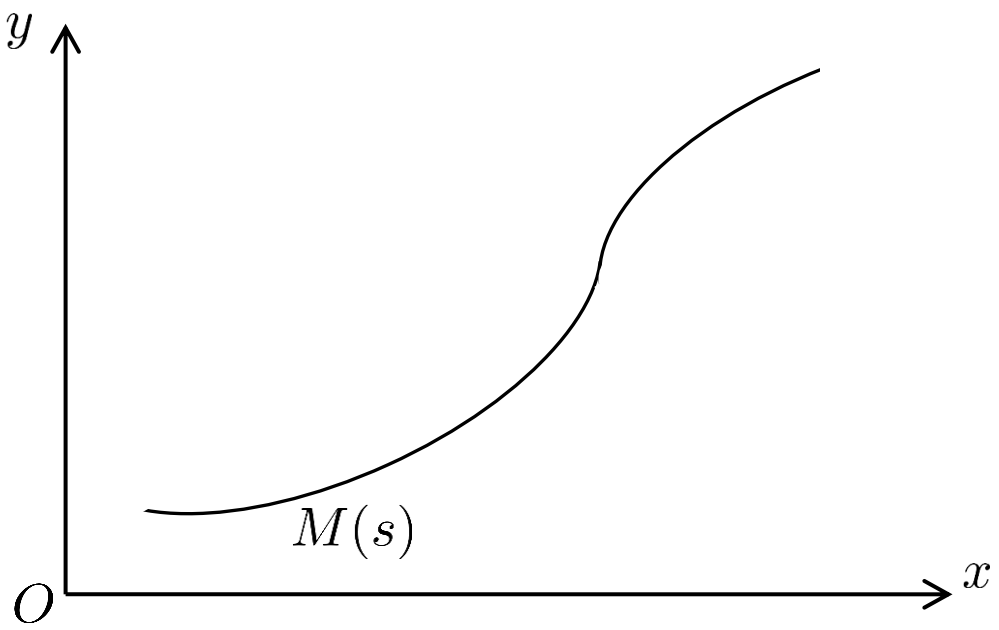
$$\omega = \frac{\mathrm{d}\xi_n}{\mathrm{d}S} \qquad \underline{t}_t = \underline{t}_0 + \omega \underline{e}_z \wedge \underline{t}_0$$

Condition de Navier-Bernoulli : $\omega = \theta$

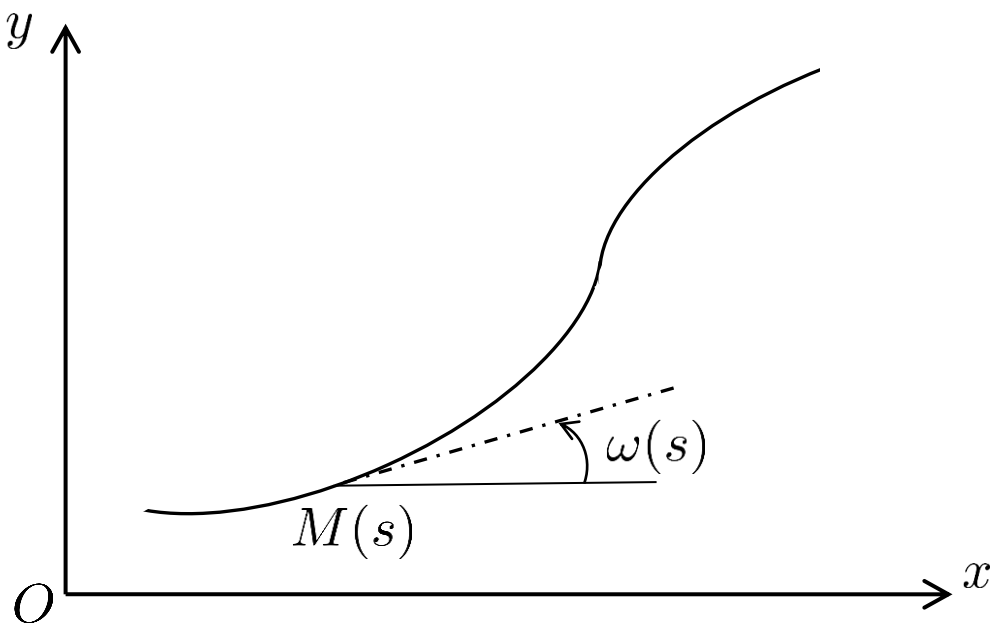
déformation de cisaillement : $\omega - \theta = \dfrac{\mathrm{d}\xi_n}{\mathrm{d}S} - \theta \overset{\text{N.B.}}{=\!=} 0$



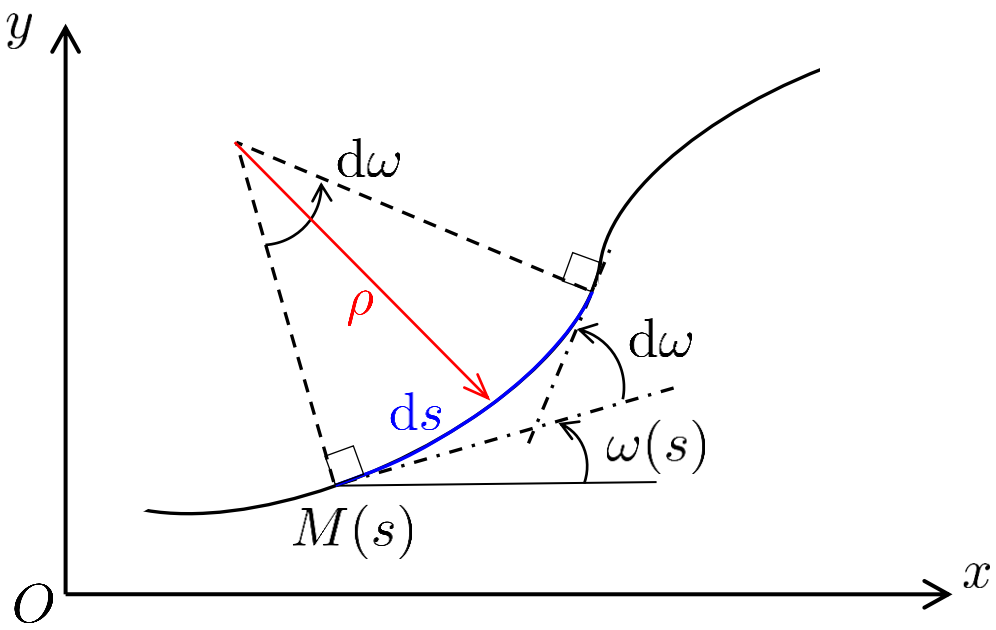
Courbure d'un arc plan



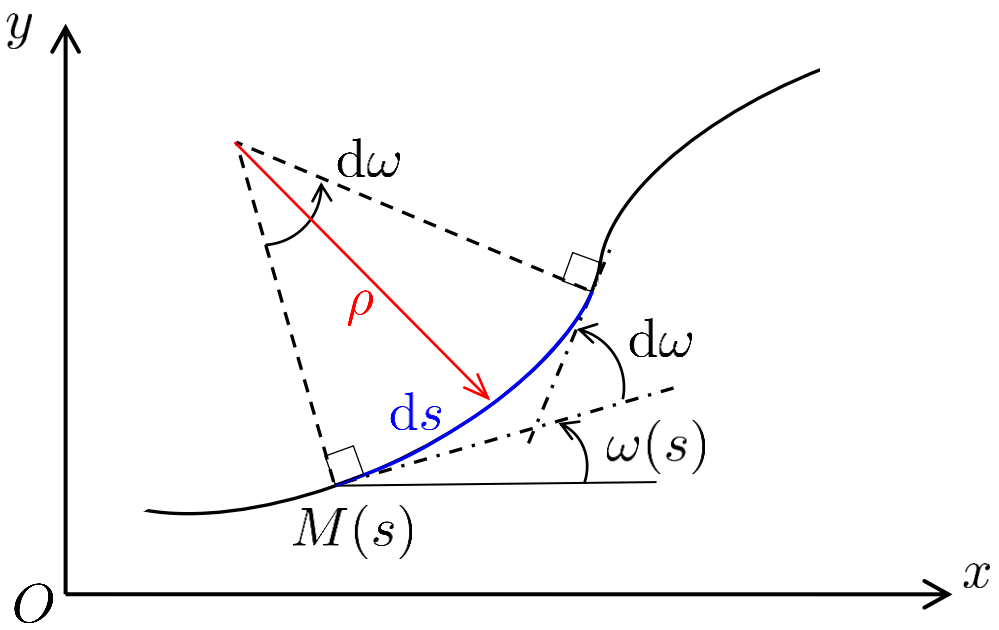
Courbure d'un arc plan



Courbure d'un arc plan

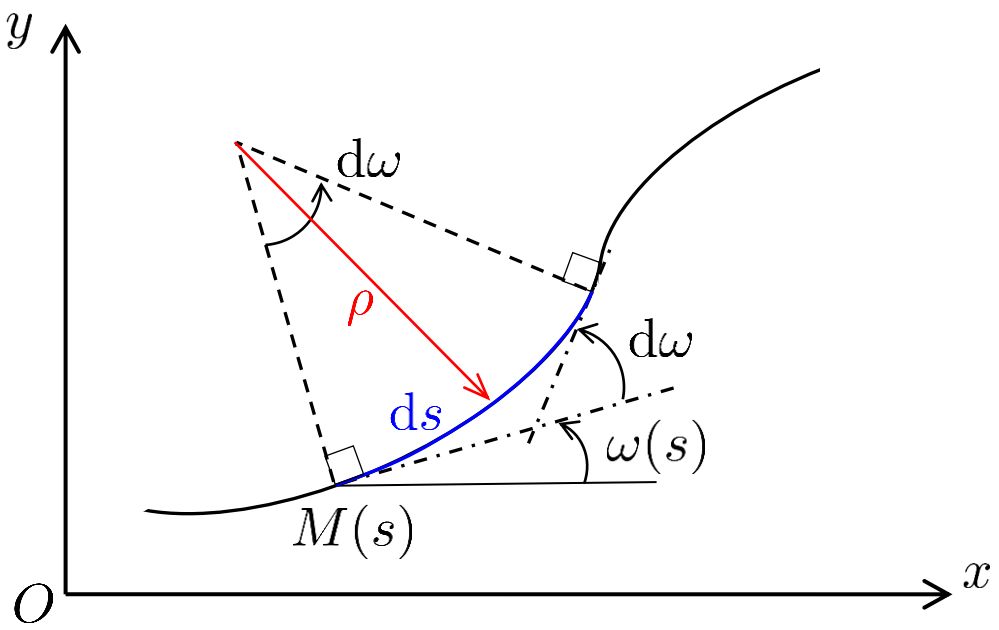


Courbure d'un arc plan



$$\rho = \frac{\mathrm{d}s}{\mathrm{d}\omega}$$

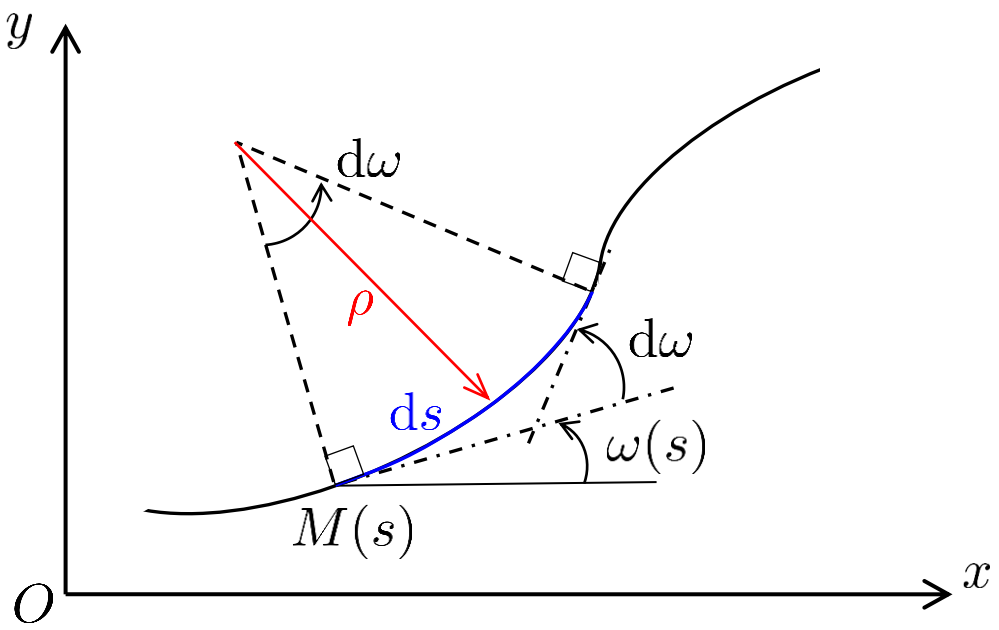
Courbure d'un arc plan



$$\rho = \frac{\mathrm{d}s}{\mathrm{d}\omega}$$

$$\chi = \frac{1}{\rho} = \frac{\mathrm{d}\omega}{\mathrm{d}s}$$

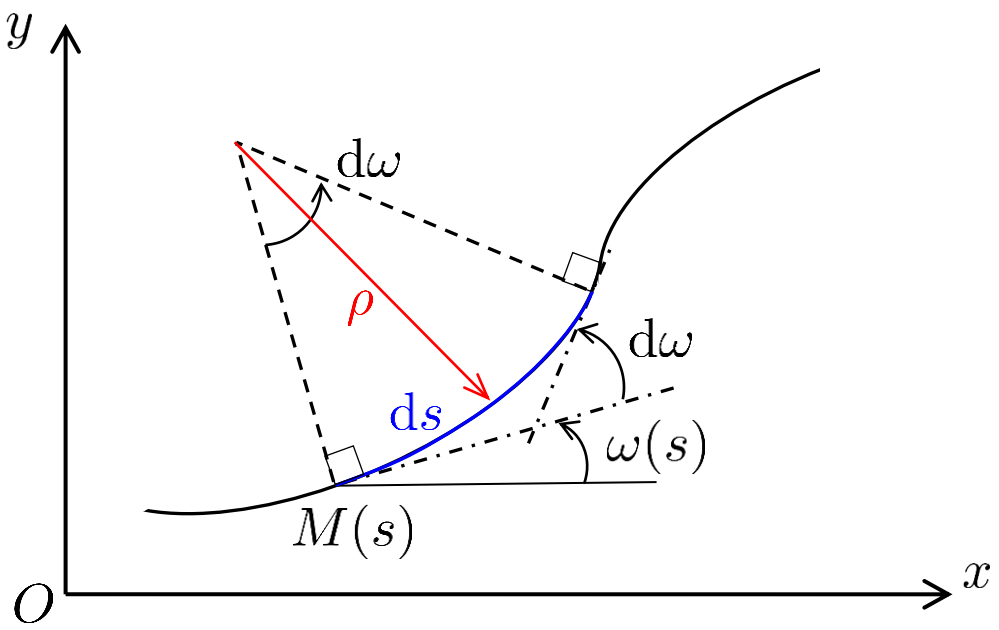
Courbure d'un arc plan



$$\rho = \frac{\mathrm{d}s}{\mathrm{d}\omega} \qquad \chi = \frac{1}{\rho} = \frac{\mathrm{d}\omega}{\mathrm{d}s}$$

$$\chi = \frac{\mathrm{d}\omega}{\mathrm{d}S}\frac{\mathrm{d}S}{\mathrm{d}s} = \frac{\mathrm{d}\omega}{\mathrm{d}S}\left(1 + \frac{\mathrm{d}\xi_t}{\mathrm{d}S}\right)^{-1}$$

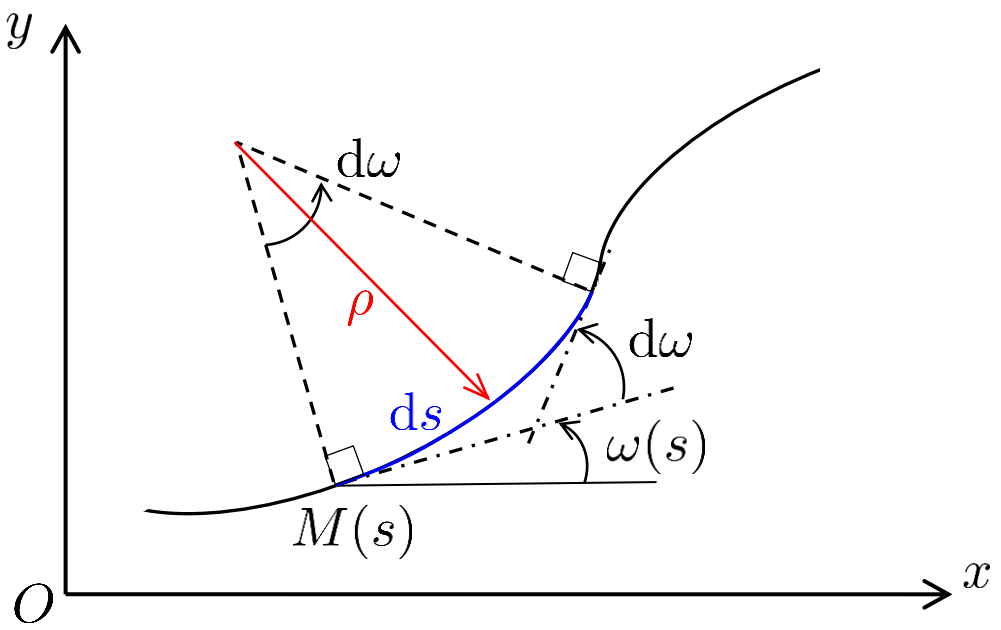
Courbure d'un arc plan



$$\rho = \frac{\mathrm{d}s}{\mathrm{d}\omega} \qquad \chi = \frac{1}{\rho} = \frac{\mathrm{d}\omega}{\mathrm{d}s}$$

$$\chi = \frac{\mathrm{d}\omega}{\mathrm{d}S}\frac{\mathrm{d}S}{\mathrm{d}s} = \frac{\mathrm{d}\omega}{\mathrm{d}S}\left(1 + \frac{\mathrm{d}\xi_t}{\mathrm{d}S}\right)^{-1} \approx \frac{\mathrm{d}\omega}{\mathrm{d}S}$$

Courbure d'un arc plan

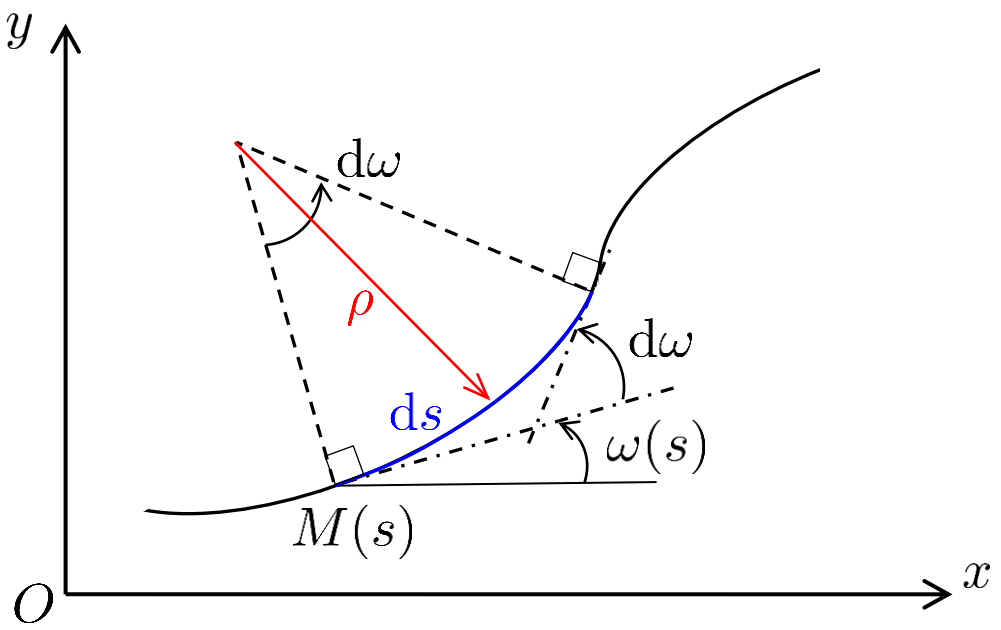


$$\rho = \frac{\mathrm{d}s}{\mathrm{d}\omega} \qquad \chi = \frac{1}{\rho} = \frac{\mathrm{d}\omega}{\mathrm{d}s}$$

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déformation de courbure : $\chi = \dfrac{\mathrm{d}\omega}{\mathrm{d}S} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}S^2}$

Courbure d'un arc plan



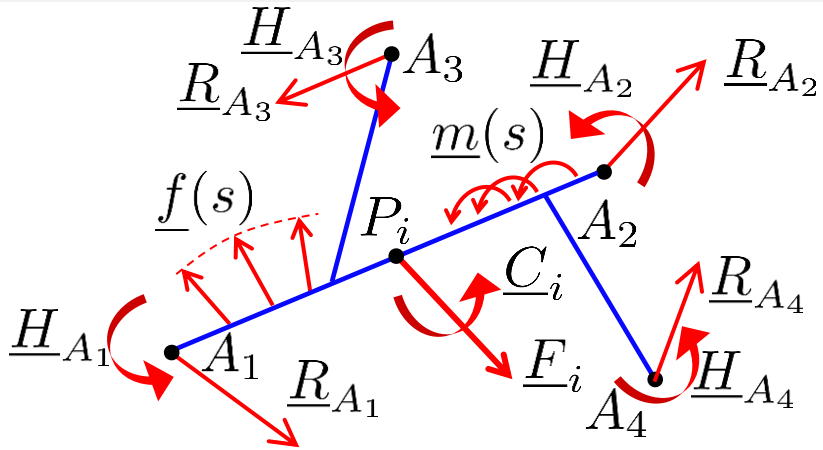
$$\rho = \frac{\mathrm{d}s}{\mathrm{d}\omega} \qquad \chi = \frac{1}{\rho} = \frac{\mathrm{d}\omega}{\mathrm{d}s}$$

$$\chi = \frac{\mathrm{d}\omega}{\mathrm{d}S}\frac{\mathrm{d}S}{\mathrm{d}s} = \frac{\mathrm{d}\omega}{\mathrm{d}S}\left(1 + \frac{\mathrm{d}\xi_t}{\mathrm{d}S}\right)^{-1} \approx \frac{\mathrm{d}\omega}{\mathrm{d}S}$$

déformation de courbure : $\chi = \dfrac{\mathrm{d}\omega}{\mathrm{d}S} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}S^2}$ N.B. $\dfrac{\mathrm{d}\theta}{\mathrm{d}S}$

Enoncé du **T**héorème des **P**uissances **V**irtuelles

$$\text{équilibre} \iff \begin{cases} \forall \text{ mvt. virtuel} \\ \\ P_{ext} = -P_{int} \end{cases}$$



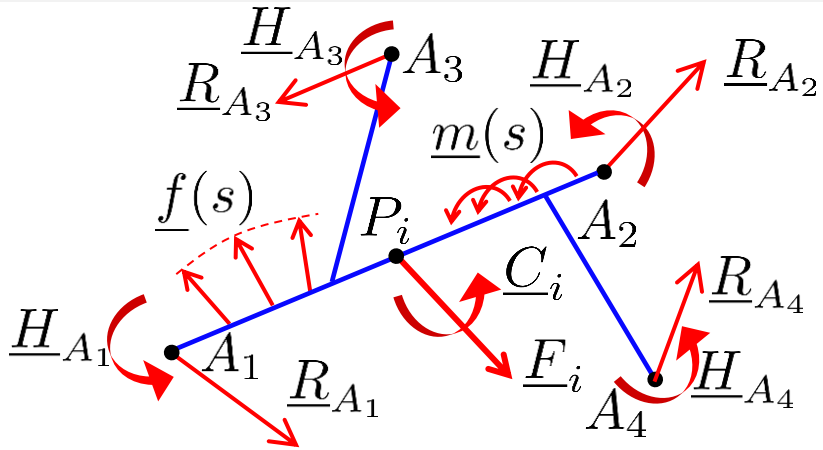
structure plane + chargement plan

$$P_{ext}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = \sum_i \left(\underline{R}_{A_i} \cdot \dot{\hat{\underline{\xi}}}(s_{A_i}) + M(s_{A_i}) \dot{\hat{\theta}}(s_{A_i}) \right) + \sum_i \left(\underline{F}_i \cdot \dot{\hat{\underline{\xi}}}(s_{P_i}) + C_i \dot{\hat{\theta}}(s_{P_i}) \right)$$
$$+ \int_{struc.} \underline{f}(s) \cdot \dot{\hat{\underline{\xi}}} + m(s) \dot{\hat{\theta}} \, \mathrm{d}s$$

$$P_{int}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = - \int_{struc.} N(s) \frac{\mathrm{d}\dot{\hat{\xi}}_t}{\mathrm{d}s}(s) + V(s) \left(\frac{\mathrm{d}\dot{\hat{\xi}}_n}{\mathrm{d}s}(s) - \dot{\hat{\theta}} \right) + M(s) \frac{\mathrm{d}\dot{\hat{\theta}}}{\mathrm{d}s}(s) \, \mathrm{d}s$$

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structure plane + chargement plan

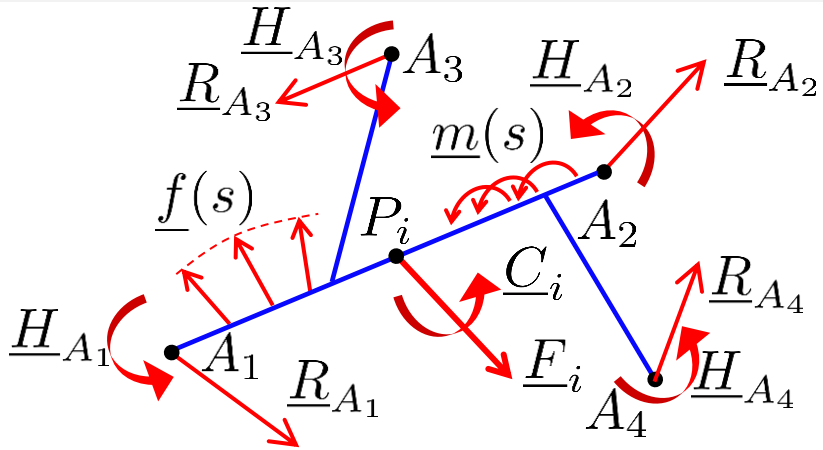
$$P_{ext}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = \sum_i \left(\underline{R}_{A_i} \cdot \dot{\hat{\underline{\xi}}}(s_{A_i}) + M(s_{A_i}) \dot{\hat{\theta}}(s_{A_i}) \right) + \sum_i \left(\underline{F}_i \cdot \dot{\hat{\underline{\xi}}}(s_{P_i}) + C_i \dot{\hat{\theta}}(s_{P_i}) \right)$$
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$\dot{\hat{\varepsilon}}$

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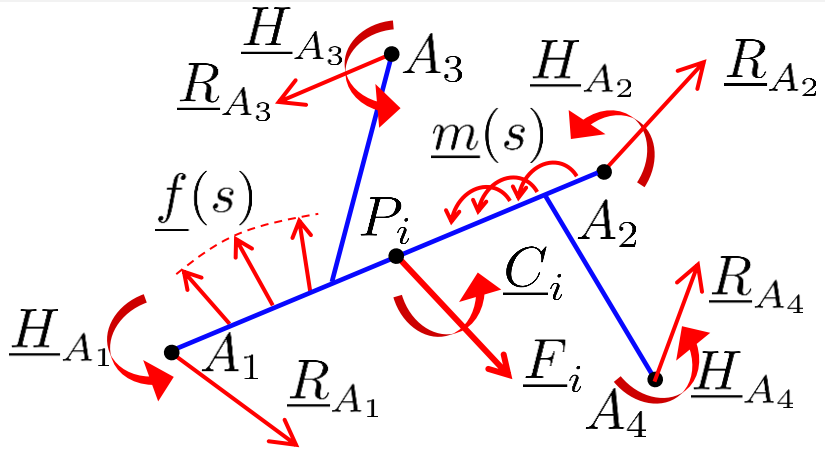
structure plane + chargement plan

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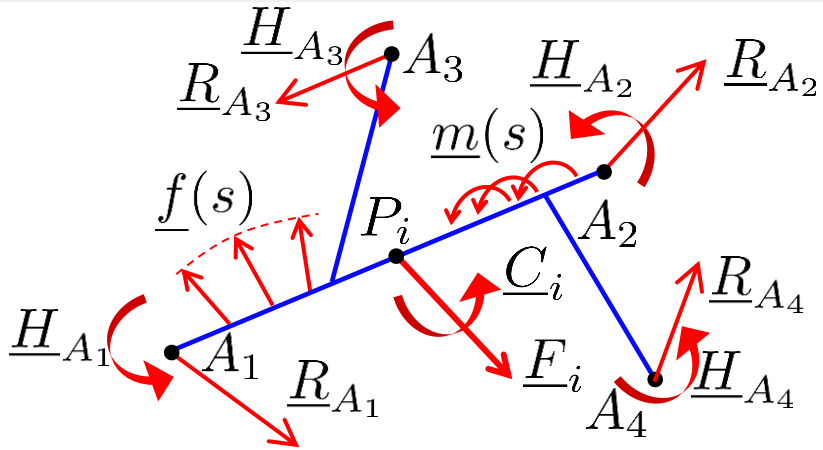
structure plane + chargement plan

$$P_{ext}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = \sum_i \left(\underline{R}_{A_i} \cdot \dot{\hat{\underline{\xi}}}(s_{A_i}) + M(s_{A_i}) \dot{\hat{\theta}}(s_{A_i}) \right) + \sum_i \left(\underline{F}_i \cdot \dot{\hat{\underline{\xi}}}(s_{P_i}) + C_i \dot{\hat{\theta}}(s_{P_i}) \right)$$
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Enoncé du **T**héorème des **P**uissances **V**irtuelles

$$\text{équilibre} \iff \begin{cases} \forall \text{ mvt. virtuel} \\ \\ P_{ext} = -P_{int} = P_{def} \end{cases}$$



structure plane + chargement plan

$$P_{ext}(\dot{\hat{\underline{\xi}}}, \dot{\hat{\theta}}) = \sum_i \left(\underline{R}_{A_i} \cdot \dot{\hat{\underline{\xi}}}(s_{A_i}) + M(s_{A_i}) \dot{\hat{\theta}}(s_{A_i}) \right) + \sum_i \left(\underline{F}_i \cdot \dot{\hat{\underline{\xi}}}(s_{P_i}) + C_i \dot{\hat{\theta}}(s_{P_i}) \right)$$
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Comportement **élastique linéaire**

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- effort normal N $\longleftrightarrow$ déformation longitudinale (ou d'extension) : $\varepsilon = \dfrac{\mathrm{d}\xi_t}{\mathrm{d}S}$

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Comportement **élastique linéaire**

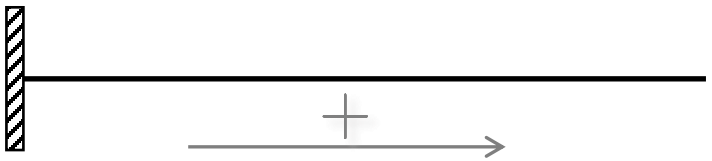
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Etat **initial** :

exemple 1:

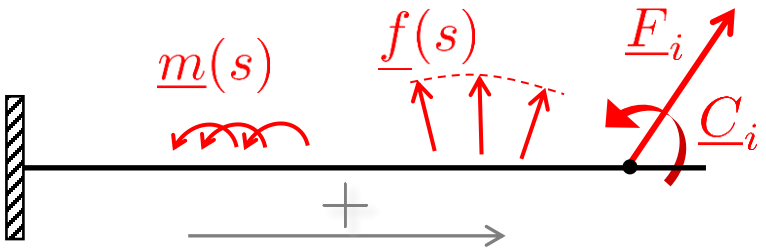


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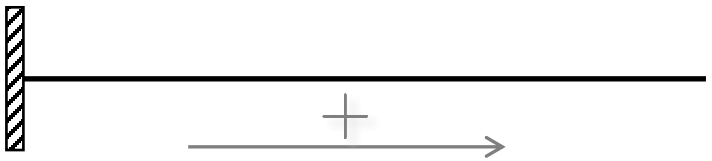


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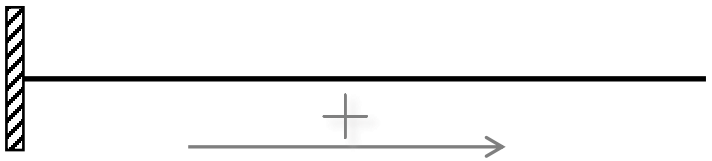


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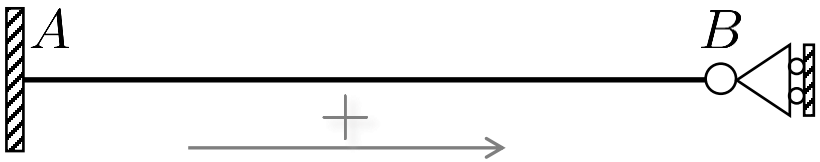
$N_0 = 0\ ;\ V_0 = 0\ ;\ M_0 = 0$

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Etat **initial** :

exemple 2:

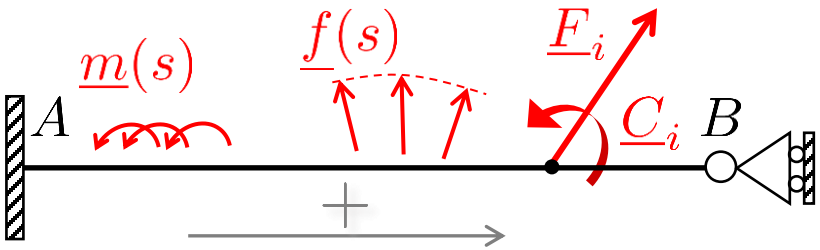


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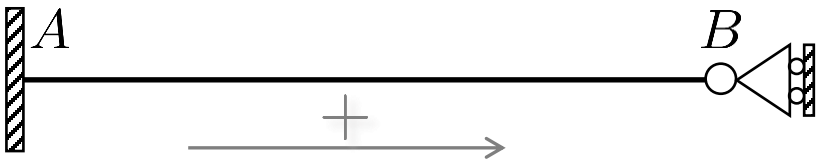


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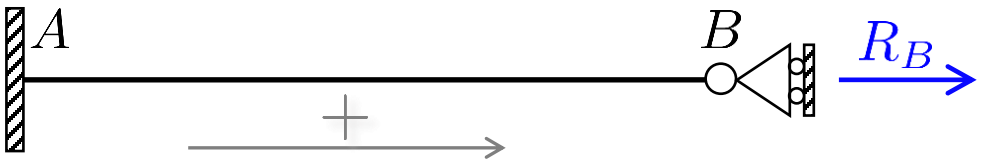


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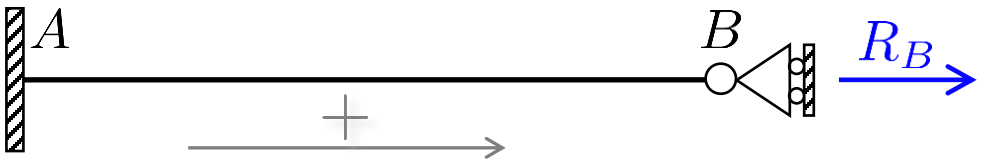


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exemple 2:



$N_0 = R_B \; ; \; V_0 = 0 \; ; \; M_0 = 0$

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- $N - N_0 = \alpha\ \varepsilon$
- $V - V_0 = \beta\ \gamma$
- $M - M_0 = \delta\ \chi$

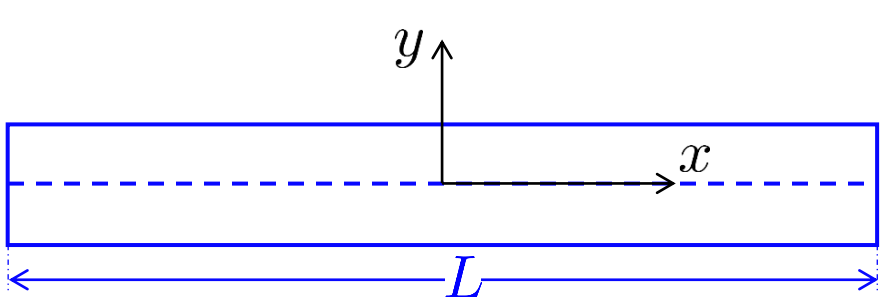
Comportement **élastique linéaire**

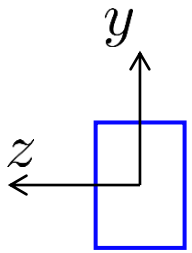
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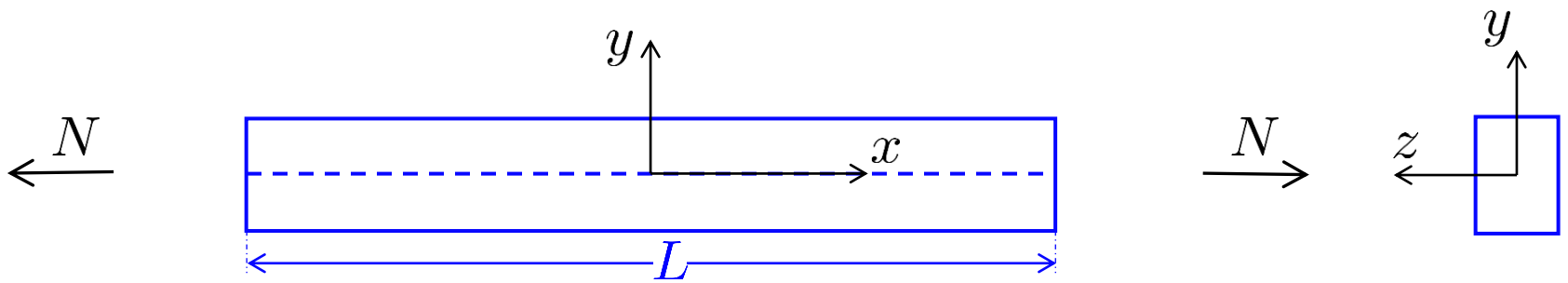
Expériences fondamentales

Traction-compression

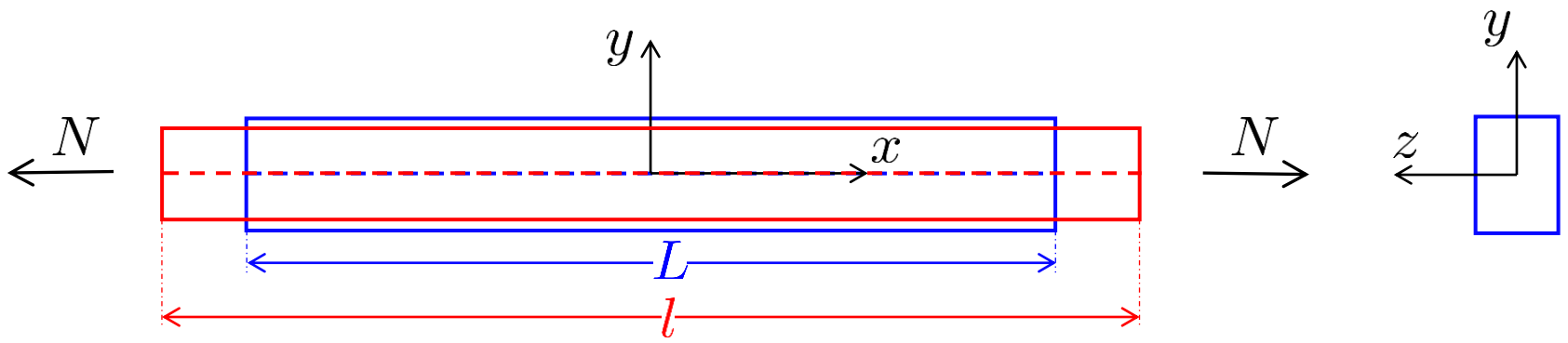




Traction-compression

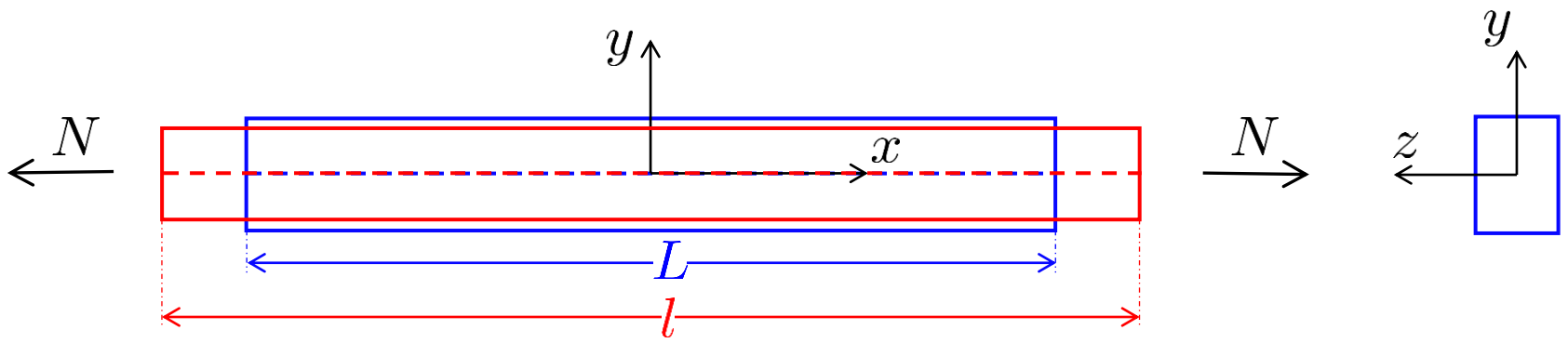


Traction-compression

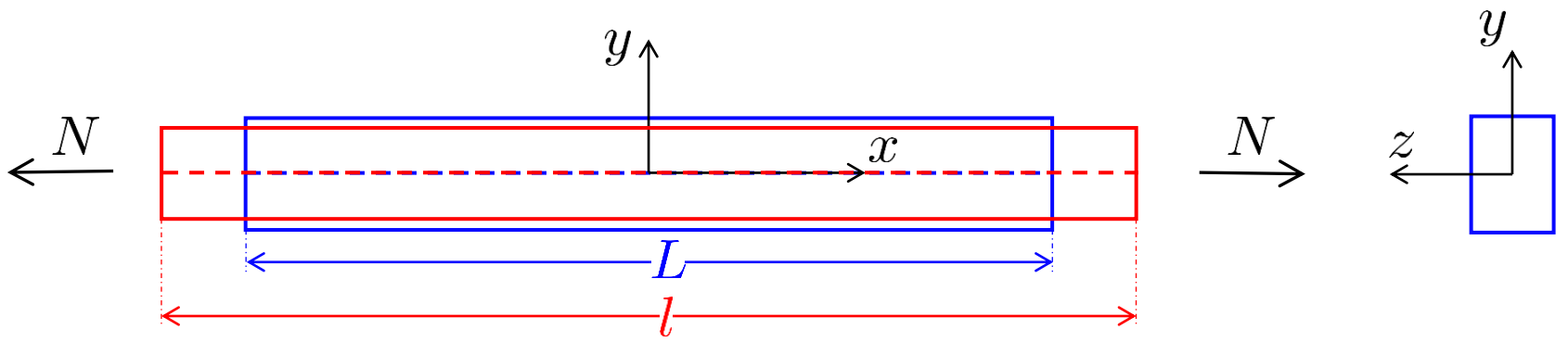


Traction-compression

$$\varepsilon = \frac{l - L}{L}$$



Traction-compression

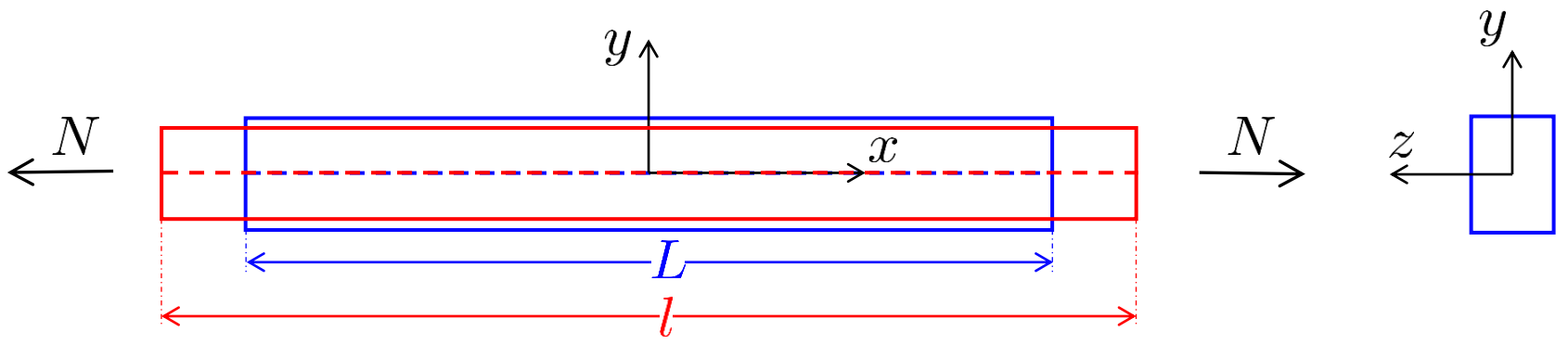


$$\varepsilon = \frac{l - L}{L} = \frac{N}{EA}$$

$$A = \int_{sect.} \mathrm{d}y\mathrm{d}z$$

Expériences fondamentales

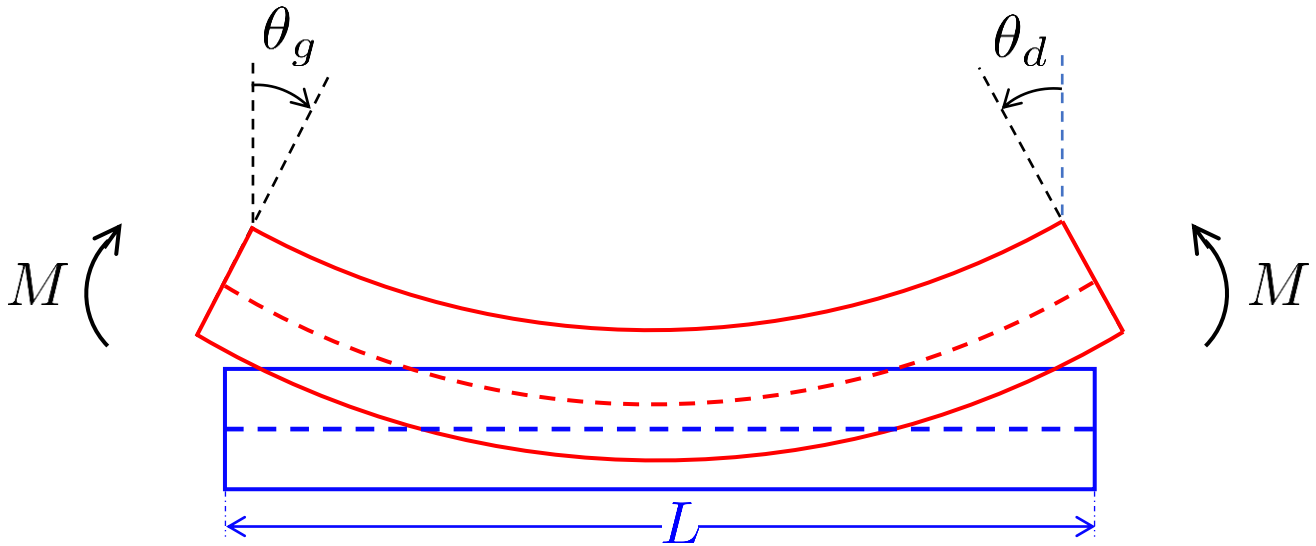
Traction-compression



$$\varepsilon = \frac{l - L}{L} = \frac{N}{EA}$$

$$A = \int_{sect.} \mathrm{d}y\mathrm{d}z$$

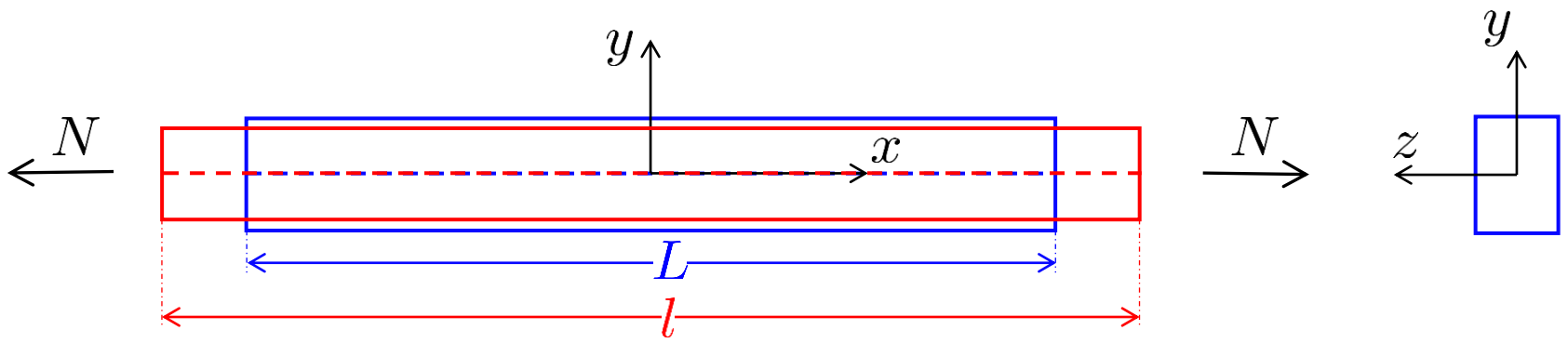
Flexion simple



Expériences fondamentales

Traction-compression

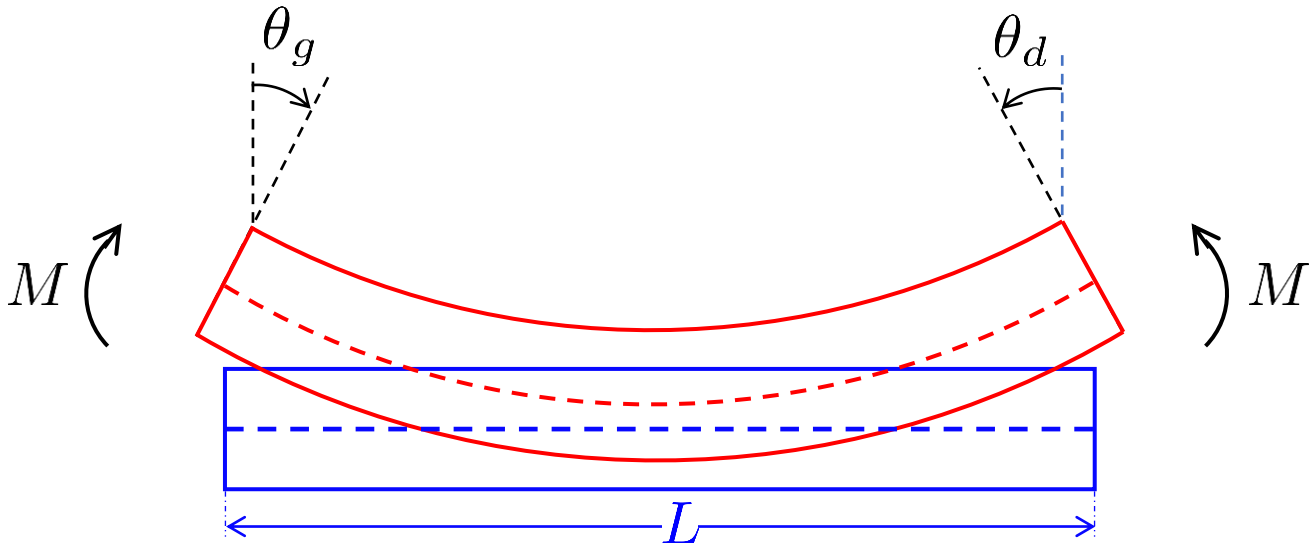
$$\varepsilon = \frac{l - L}{L} = \frac{N}{EA}$$

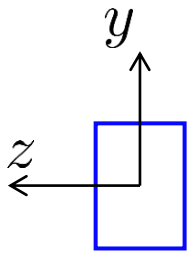


$$A = \int_{sect.} \mathrm{d}y\mathrm{d}z$$

Flexion simple

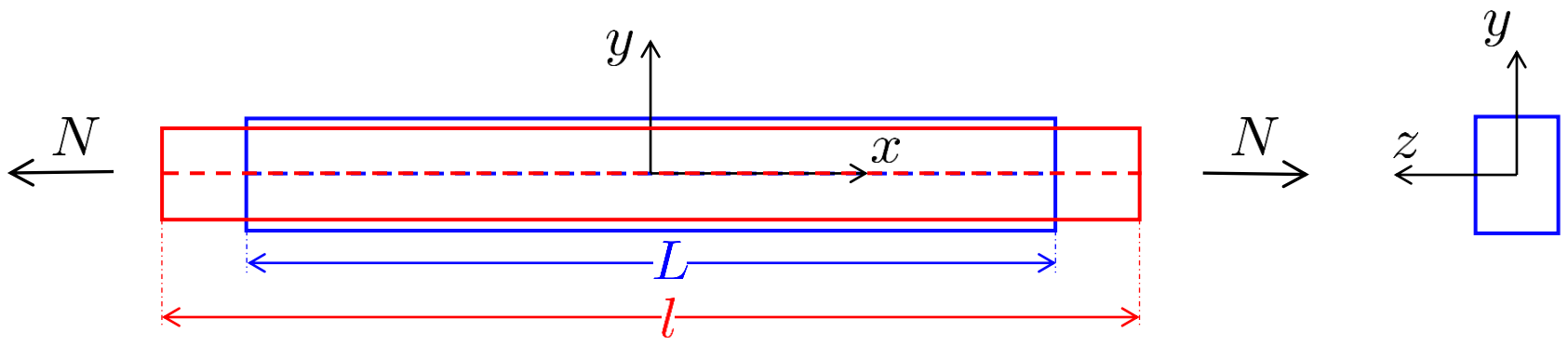
$$\chi = \frac{\theta_d - \theta_g}{L}$$





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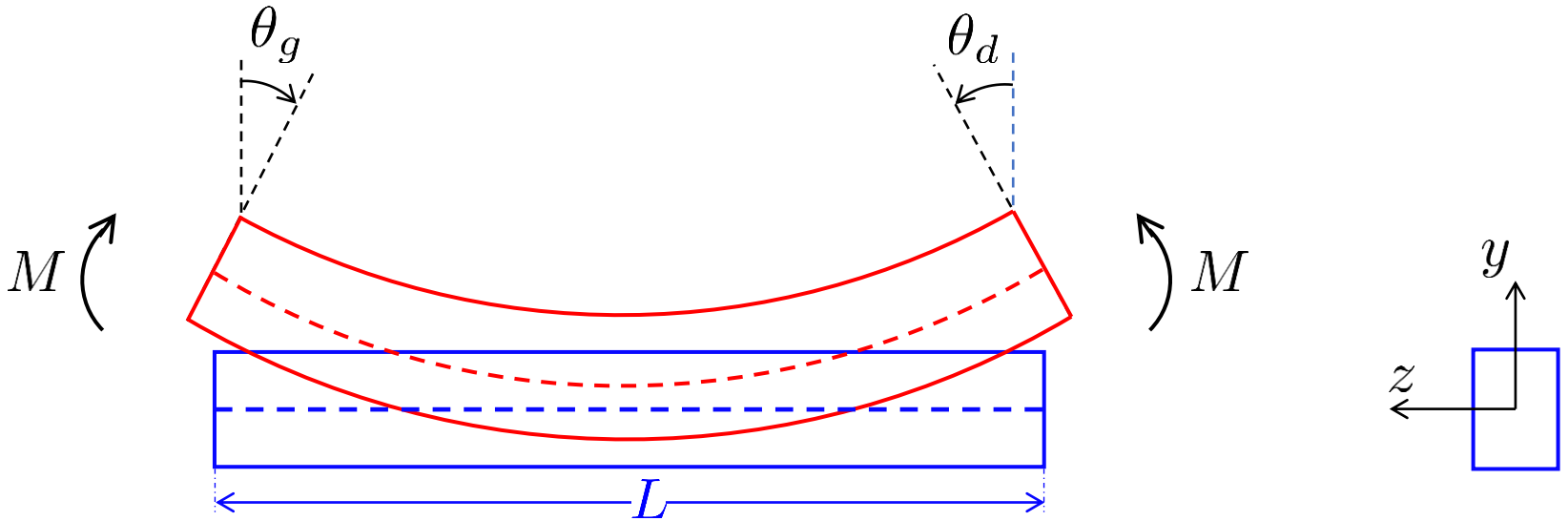
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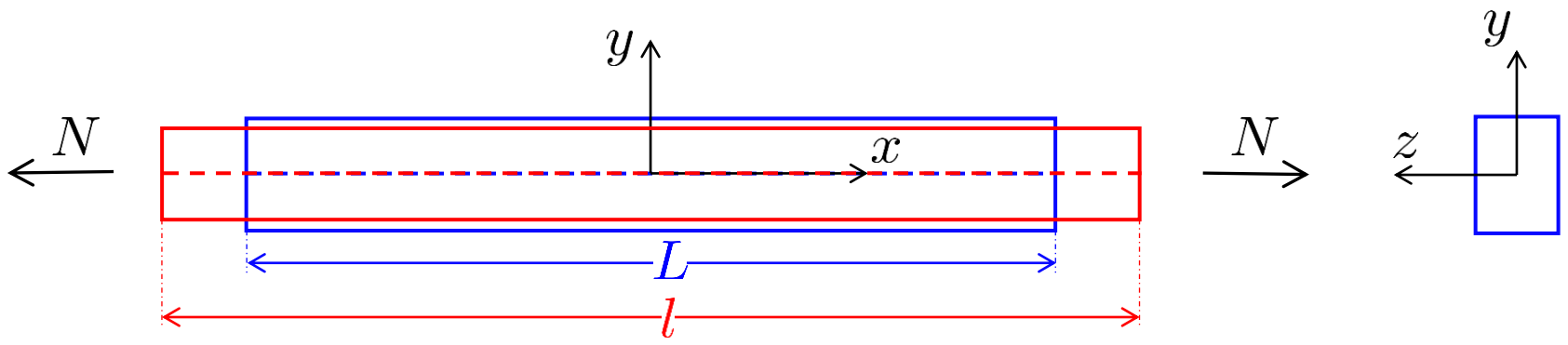
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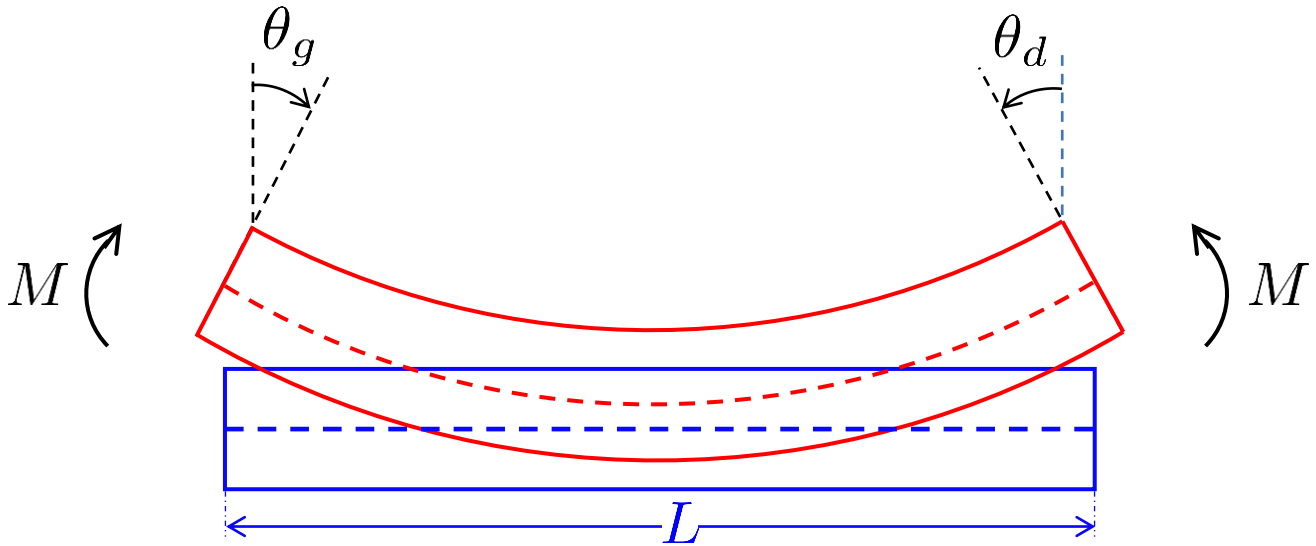
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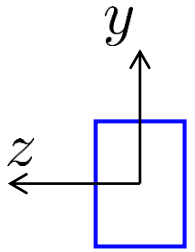


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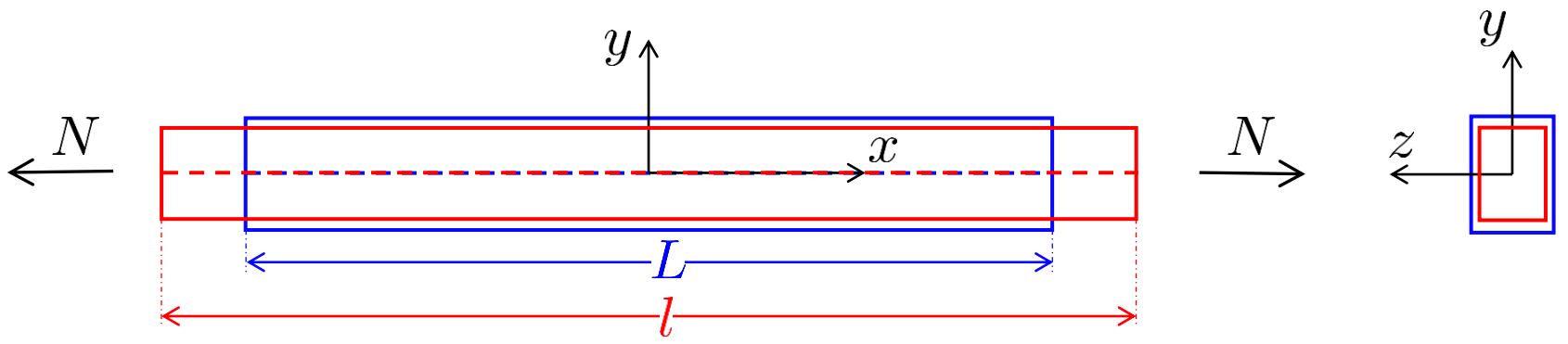


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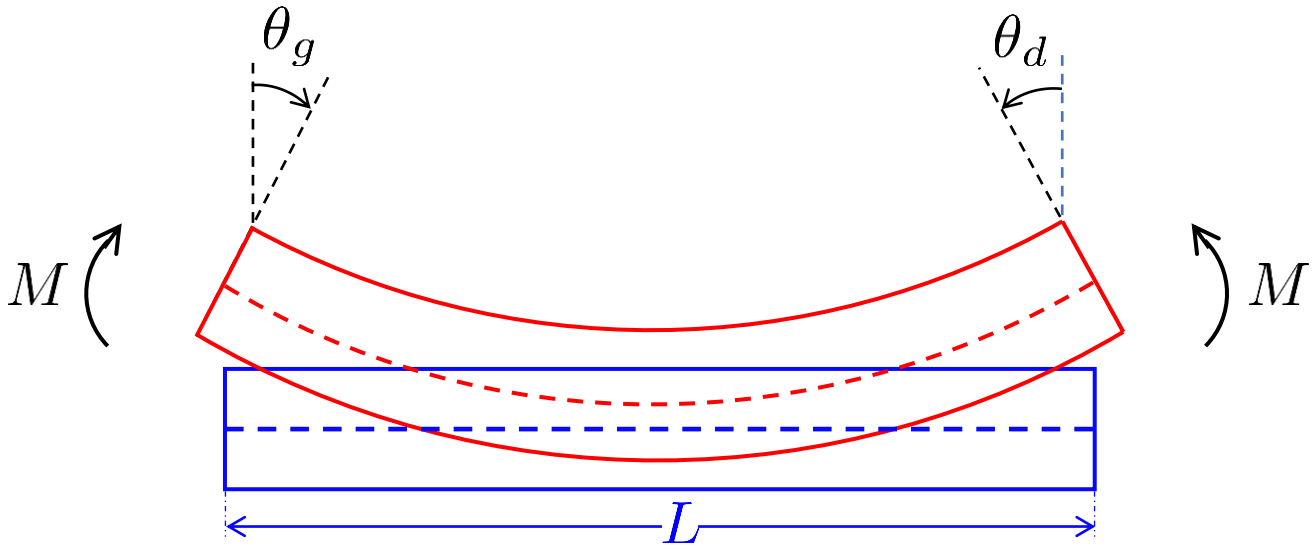
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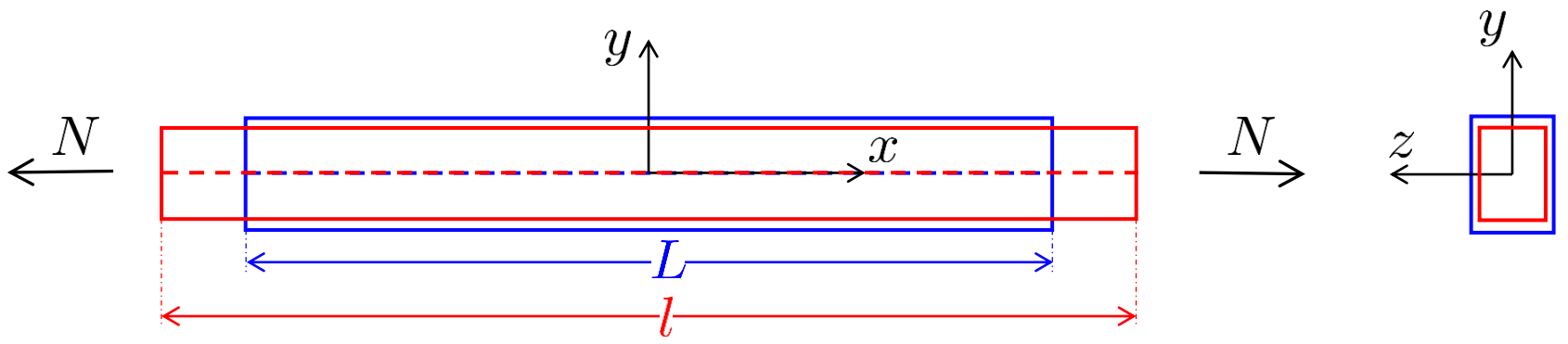


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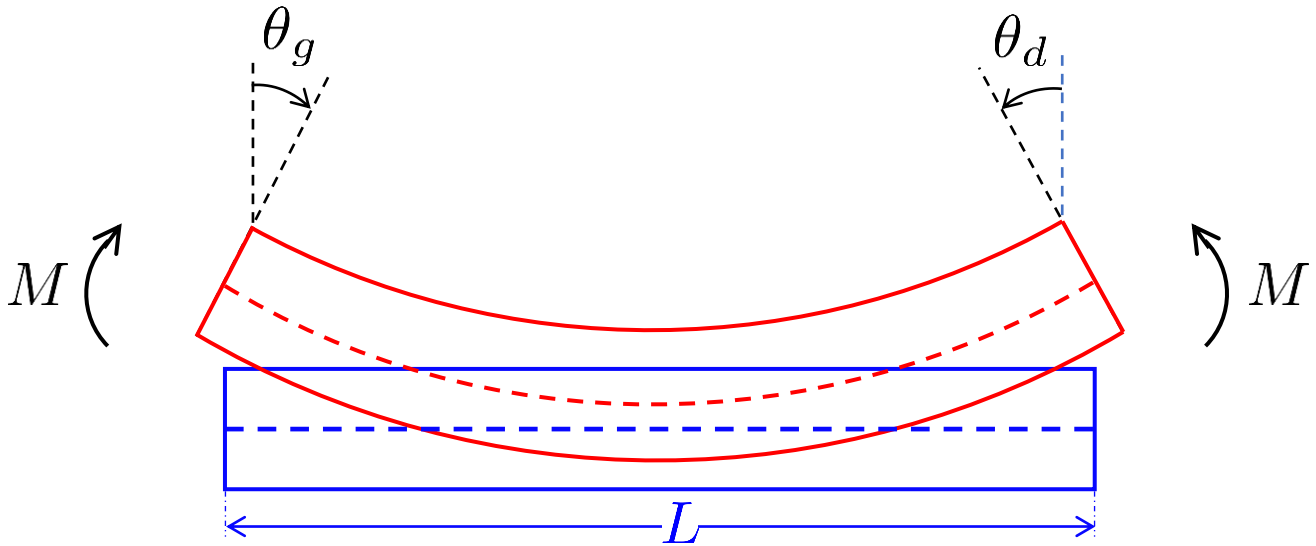
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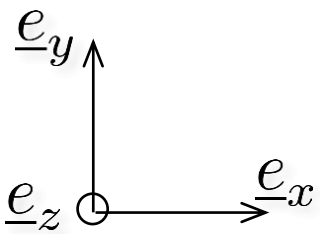


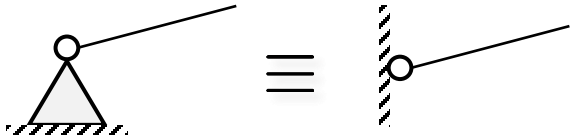
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Conditions aux appuis : exemples

Cas des structures planes chargées dans leur plan





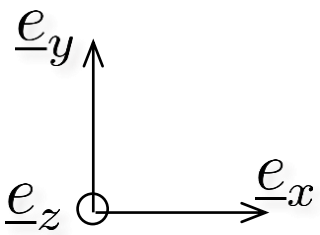
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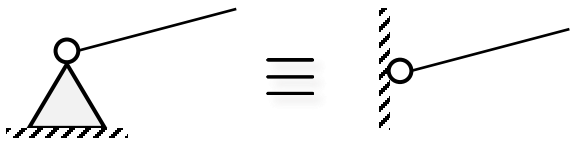
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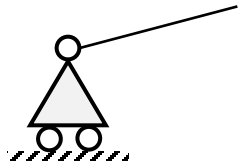




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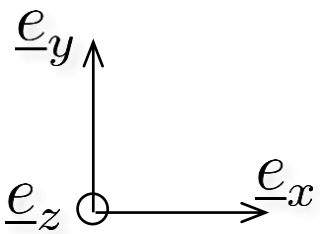


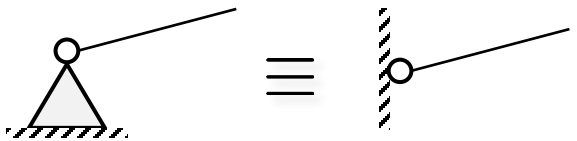
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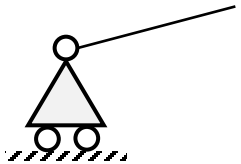




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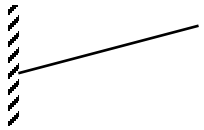
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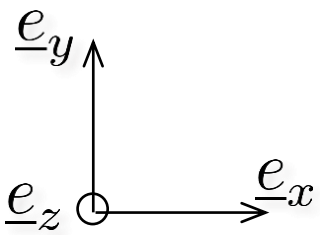


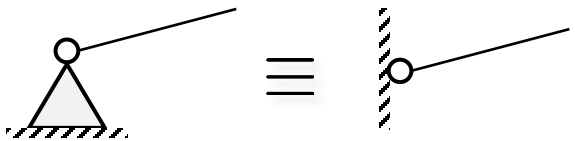
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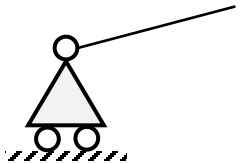




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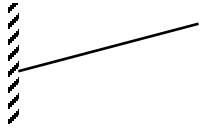
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Cadre de travail : Compléments

- structure plane
- tronçons rectilignes
- chargement plan
- cinématique plane
- transformation infinitésimale : $|\theta|, \left|\frac{\mathrm{d}\xi_t}{\mathrm{d}S}\right|, \left|\frac{\mathrm{d}\xi_n}{\mathrm{d}S}\right| \ll 1$

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Hypothèse des Petites Perturbations

H.P.P

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Données et solution d'un problème d'élastostatique HPP

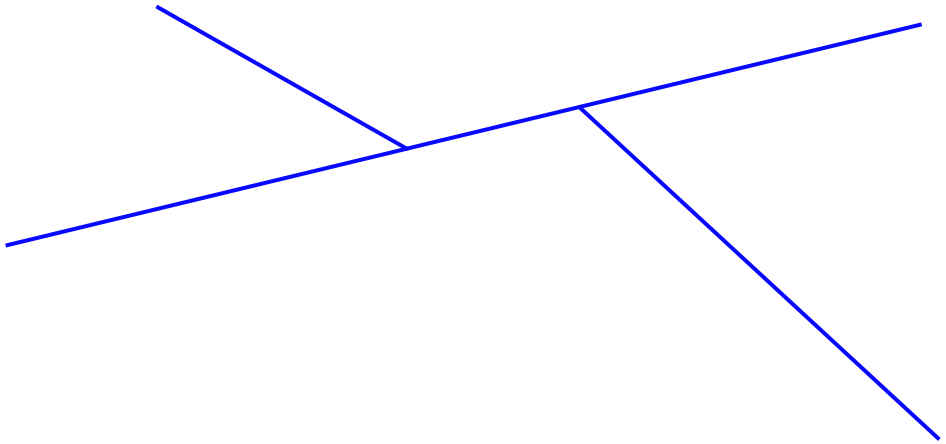
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Données et solution d'un problème d'élastostatique HPP

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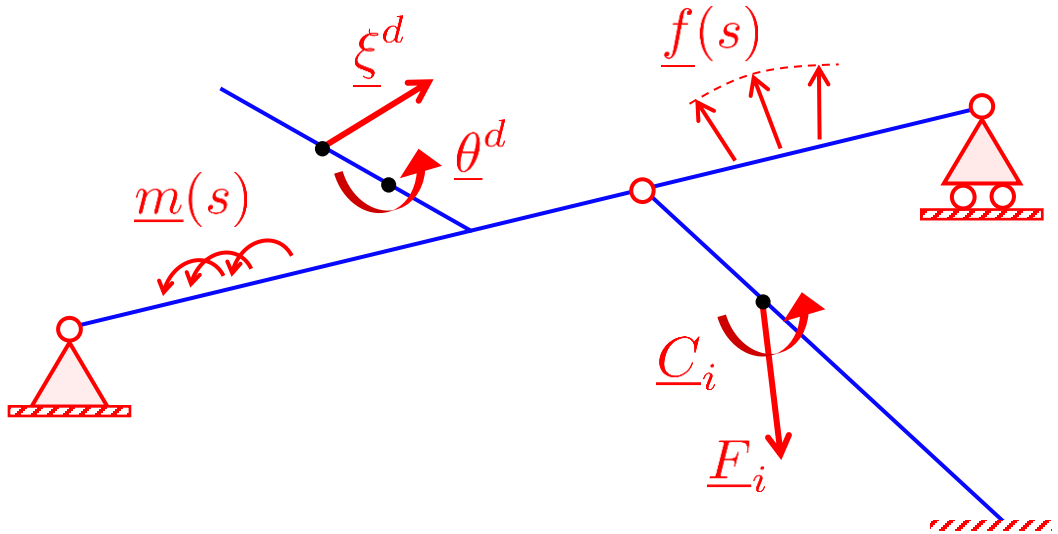


Données et solution d'un problème d'élastostatique HPP

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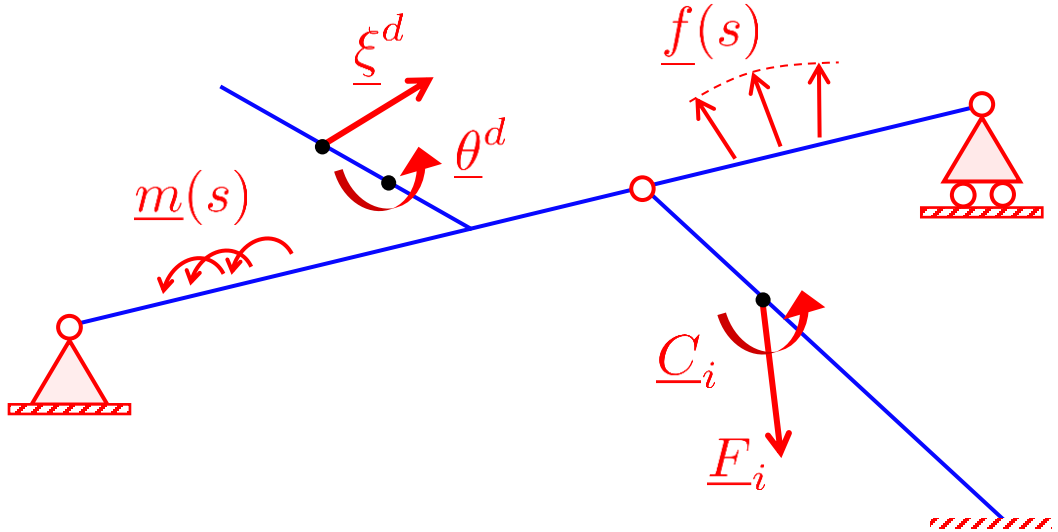
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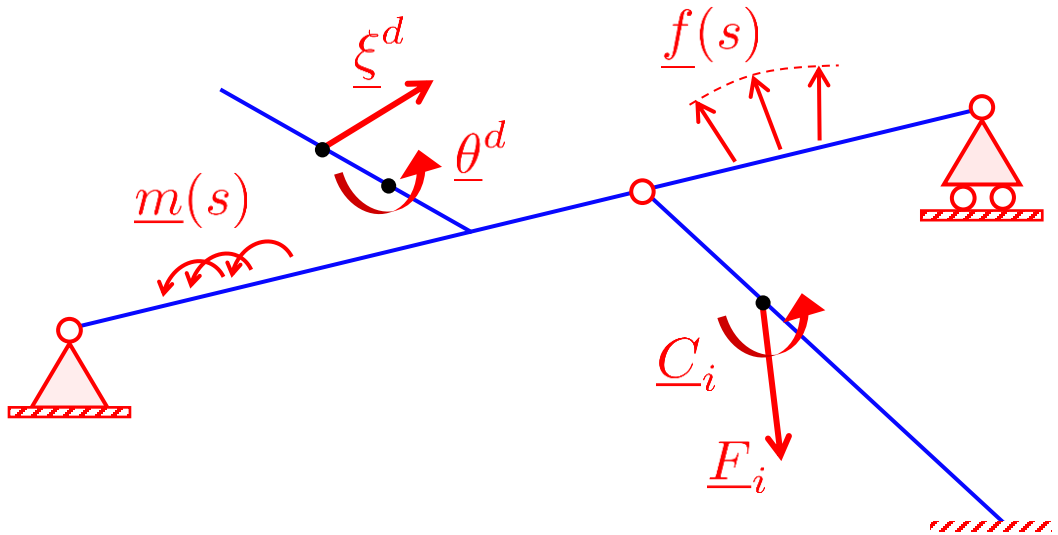
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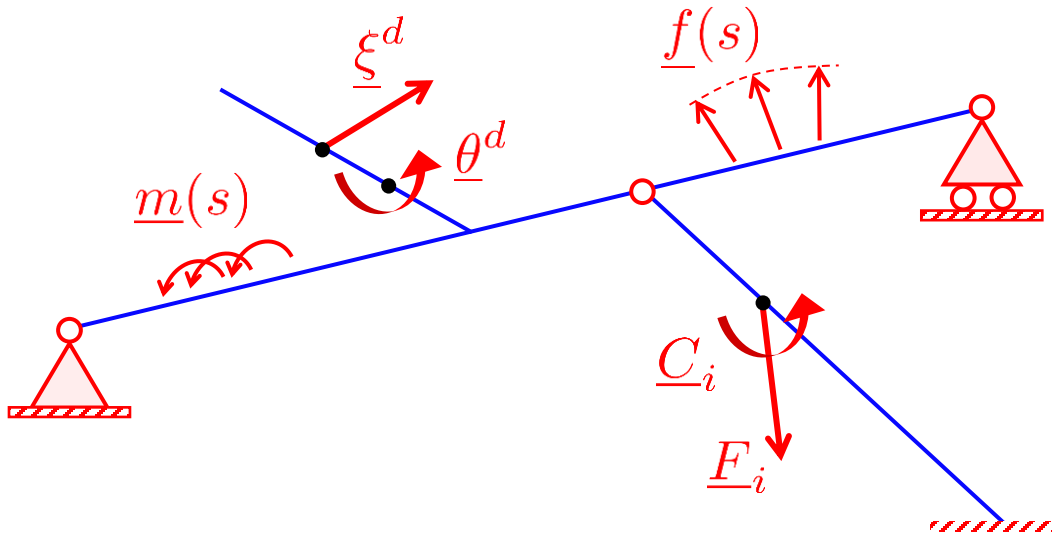
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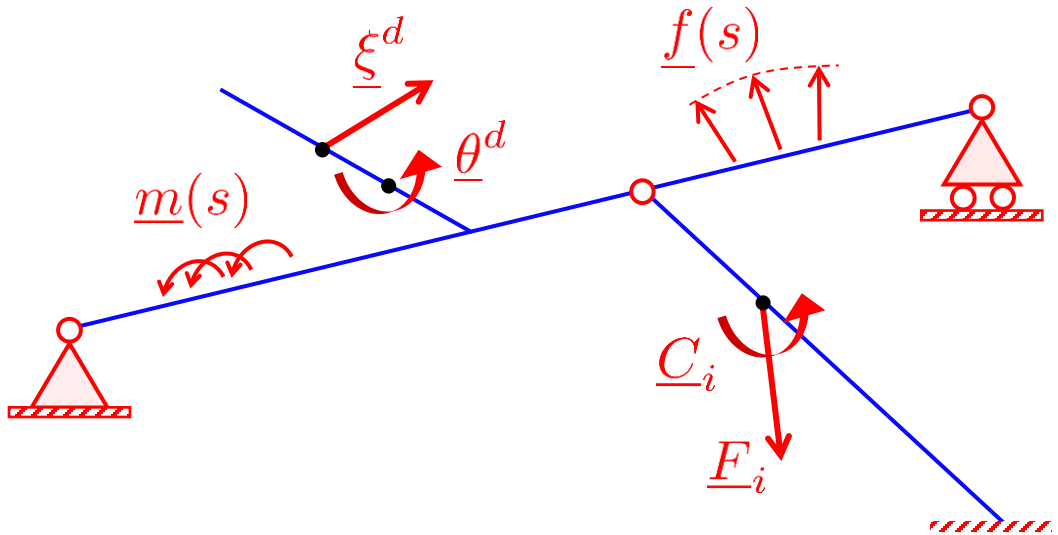
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Solution d'un problème d'élastostatique

$(\underline{\xi}, \theta)$: réguliers + C.L. $\qquad (N, V, M)$

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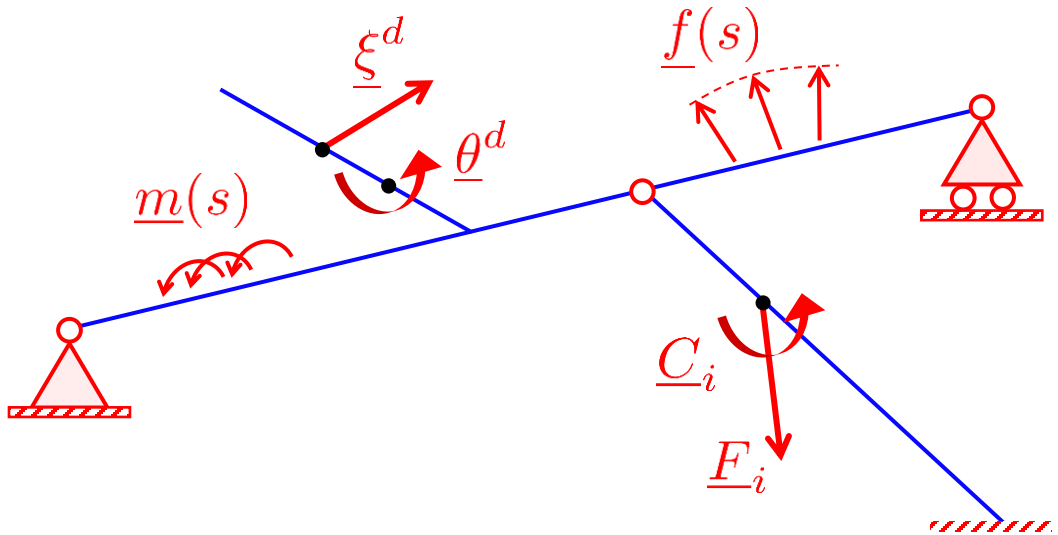
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Solution d'un problème d'élastostatique

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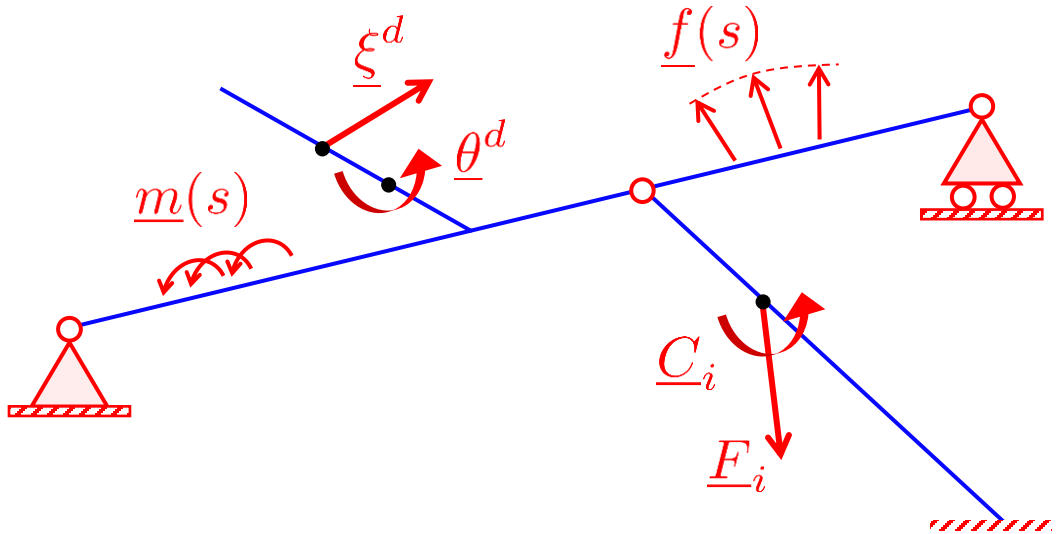
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$\updownarrow$

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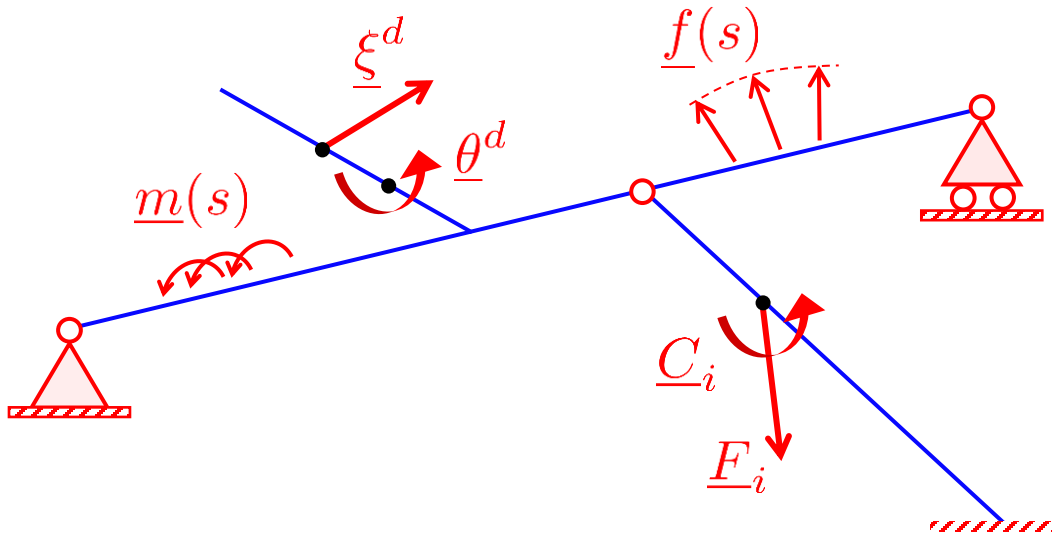
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$\longleftrightarrow$

$$N = N_0 + EA\,\varepsilon$$
$$M = M_0 + EI_z\,\chi$$

Méthode de résolution directe

problème de degré d'hyperstaticité $p : (Y_k), k \in \{1, ...p\}$

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problème de degré d'hyperstaticité p : $(Y_k), k \in \{1, ...p\}$

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ξ_t et θ : $\mathcal{C}_0$ et $\mathcal{C}_1^{pm}$; ξ_n : $\mathcal{C}_0, \mathcal{C}_1$ et $\mathcal{C}_2^{pm}$

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comportement élastique linéaire
$$\begin{cases} \dfrac{N - N_0}{EA} = \dfrac{\mathrm{d}\xi_t}{\mathrm{d}S} \\ \dfrac{M - M_0}{EI_z} = \dfrac{\mathrm{d}\theta}{\mathrm{d}S} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}S^2} \end{cases}$$

ξ_t et θ : $\mathcal{C}_0$ et $\mathcal{C}_1^{pm}$; ξ_n : $\mathcal{C}_0, \mathcal{C}_1$ et $\mathcal{C}_2^{pm}$

intégration $\Longrightarrow$
$$\begin{cases} \xi_t(Y_k, S) \\ \xi_n(Y_k, S) \end{cases}$$

Méthode de résolution directe

problème de degré d'hyperstaticité p : $(Y_k), k \in \{1, ...p\}$

statique
$$\begin{cases} N = N(\underline{Q}, Y_k, S) \\ V = V(\underline{Q}, Y_k, S) \\ M = M(\underline{Q}, Y_k, S) \end{cases}$$

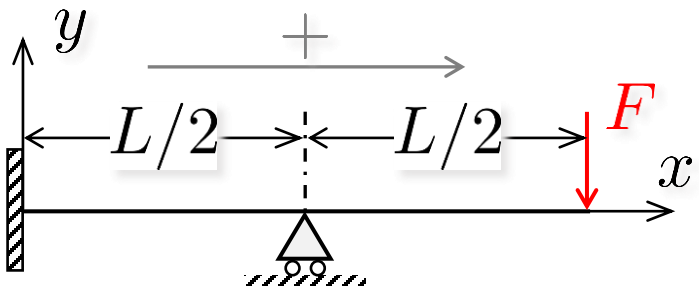
comportement élastique linéaire
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ξ_t et θ : $\mathcal{C}_0$ et $\mathcal{C}_1^{pm}$; ξ_n : $\mathcal{C}_0, \mathcal{C}_1$ et $\mathcal{C}_2^{pm}$

intégration $\Longrightarrow$ $\begin{cases} \xi_t(Y_k, S) \\ \xi_n(Y_k, S) \end{cases}$ $\xrightarrow[\text{conditions aux appuis}]{\text{régularité de } \xi_t, \xi_n \text{ et } \theta}$ Solution du pb $\begin{cases} Y_k \longrightarrow (N, V, M) \\ \xi_t(S) \\ \xi_n(S) \end{cases}$

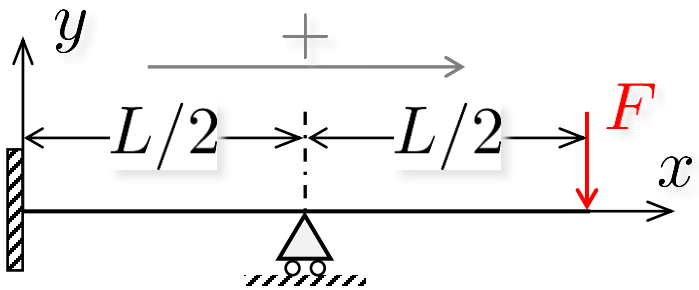
Exemple

Etat initial naturel



Exemple

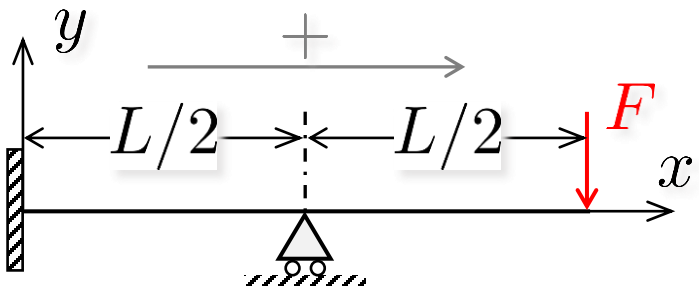
Etat initial naturel



Statique :

Exemple

Etat initial naturel

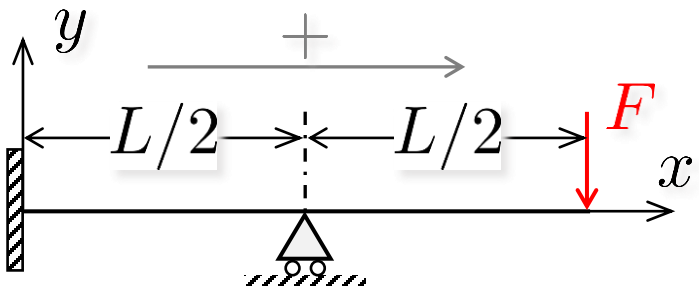


Statique :

$x > L/2$:

Exemple

Etat initial naturel

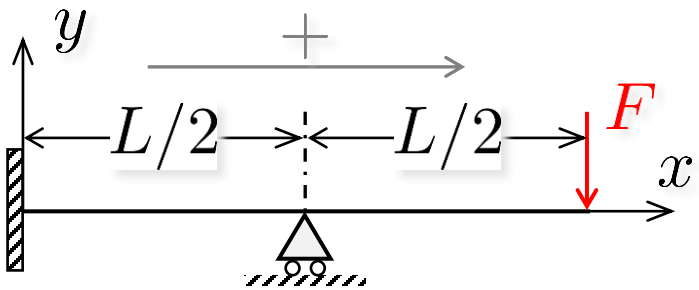


Statique :

$$x > L/2 : \; N = 0 \, ; \, V = -F \, ; \, M = -F(L - x)$$

Exemple

Etat initial naturel



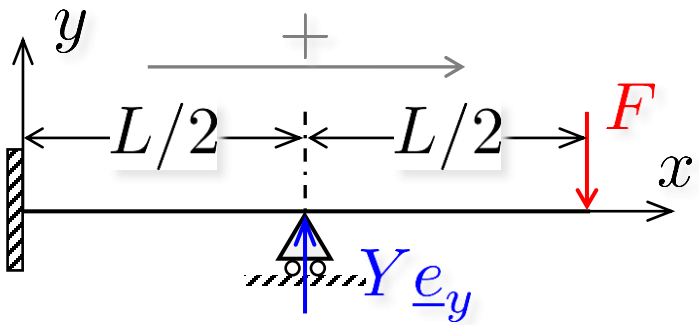
Statique :

$x > L/2$: $N = 0\,;\, V = -F\,;\, M = -F(L - x)$

$x < L/2$:

Exemple

Etat initial naturel



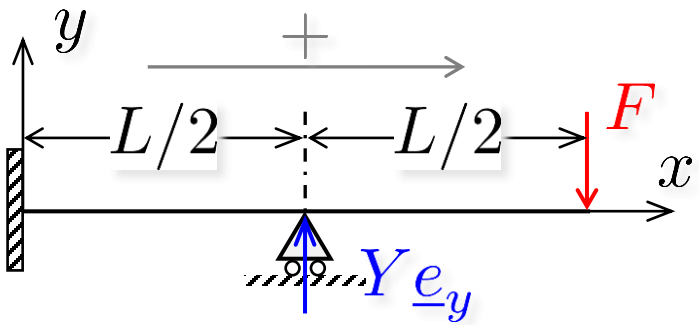
Statique :

$x > L/2 : \; N = 0 \, ; \, V = -F \, ; \, M = -F(L - x)$

$x < L/2 : \; N = 0 \, ; \, V = -F + Y \, ; \, M = -F(L - x) + Y(L/2 - x)$

Exemple

Etat initial naturel



Statique :

$$x > L/2 : \; N = 0 \, ; \, V = -F \, ; \, M = -F(L - x)$$

$$x < L/2 : \; N = 0 \, ; \, V = -F + Y \, ; \, M = -F(L - x) + Y(L/2 - x)$$

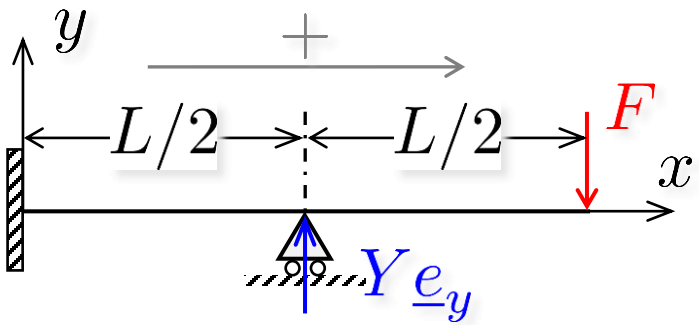
Comportement :

$$x \geq L/2 : \quad \varepsilon = \frac{N}{EA} = 0$$

$$x \leq L/2 : \quad \varepsilon = \frac{N}{EA} = 0$$

Exemple

Etat initial naturel



Statique :

$$x > L/2 : \; N = 0 \, ; \, V = -F \, ; \, M = -F(L - x)$$

$$x < L/2 : \; N = 0 \, ; \, V = -F + Y \, ; \, M = -F(L - x) + Y(L/2 - x)$$

Comportement :

$$x \geq L/2 : \quad \varepsilon = \frac{N}{EA} = 0$$

$$x \leq L/2 : \quad \varepsilon = \frac{N}{EA} = 0$$

$$\mathrm{d}S = \mathrm{d}x$$

$$\xi_t = \xi_x$$

$$\xi_n = \xi_y$$

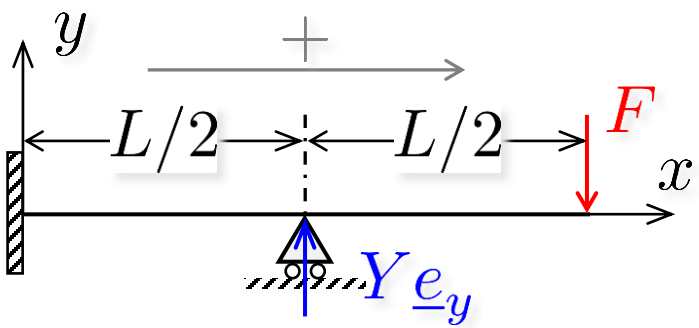
Intégration + CL :

$$x \in [0, L] : \varepsilon = 0$$

$$\rightarrow \xi_t(x) = cste.$$

Exemple

Etat initial naturel



Statique :

$$x > L/2 : \; N = 0 \, ; \, V = -F \, ; \, M = -F(L - x)$$

$$x < L/2 : \; N = 0 \, ; \, V = -F + Y \, ; \, M = -F(L - x) + Y(L/2 - x)$$

Comportement :

$$x \geq L/2 : \quad \varepsilon = \frac{N}{EA} = 0$$

$$x \leq L/2 : \quad \varepsilon = \frac{N}{EA} = 0$$

$$\mathrm{d}S = \mathrm{d}x$$

$$\xi_t = \xi_x$$

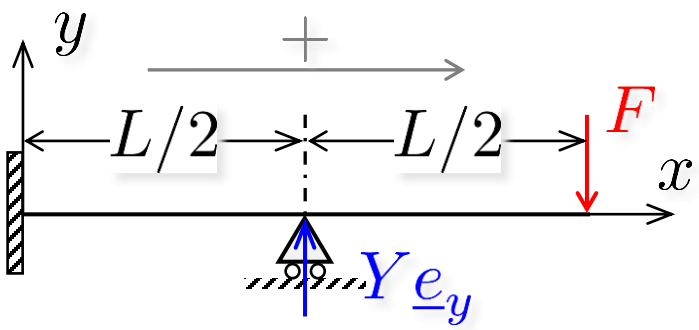
$$\xi_n = \xi_y$$

Intégration + CL :

$$x \in [0, L] : \varepsilon = 0$$

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Exemple



Statique :

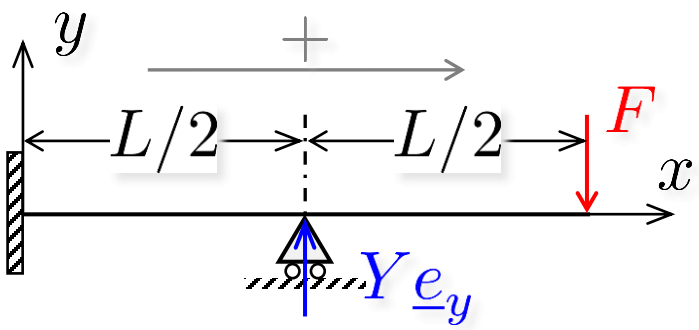
$$x \geq L/2 : \quad M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + Y(L/2-x)$$

Comportement :

$$x \leq L/2 : \quad \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x) + \frac{Y}{EI_z}(L/2-x)$$

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

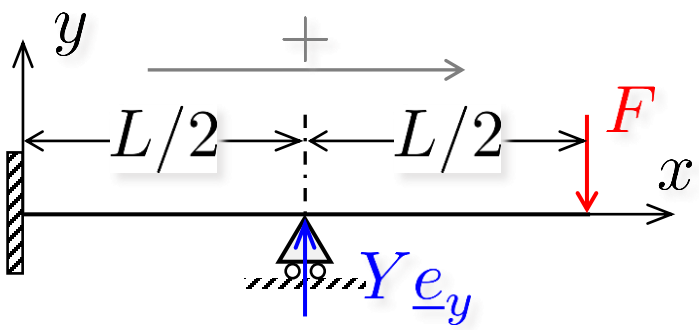
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Comportement :

$$x \leq L/2 : \quad \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x) + \frac{Y}{EI_z}(L/2-x)$$

Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + CL :

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + Y(L/2-x)$$

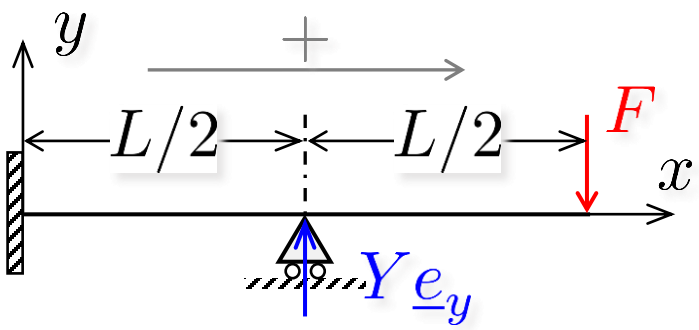
Comportement :

$$x \leq L/2 : \quad \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x) + \frac{Y}{EI_z}(L/2-x)$$

Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + CL :

$x \leq L/2$:

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + Y(L/2-x)$$

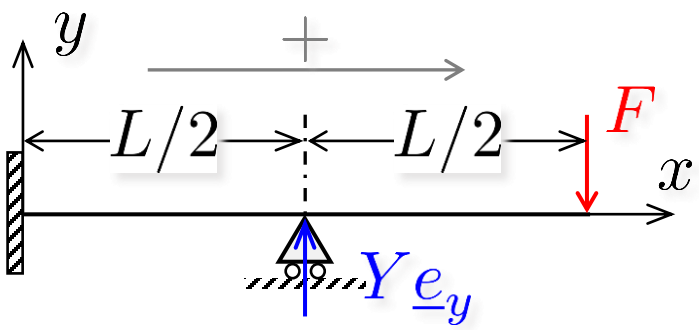
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Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + CL :

$$x \leq L/2 : \theta(x) = -F(Lx - x^2/2)/(EI_z) + Y(Lx/2 - x^2/2)/(EI_z) + \text{cste.}$$

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

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Comportement :

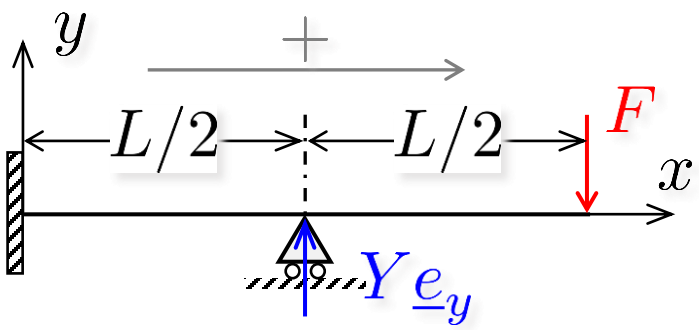
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Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + CL :

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$= 0 \, (\theta(0) = 0)$

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + Y(L/2-x)$$

Comportement :

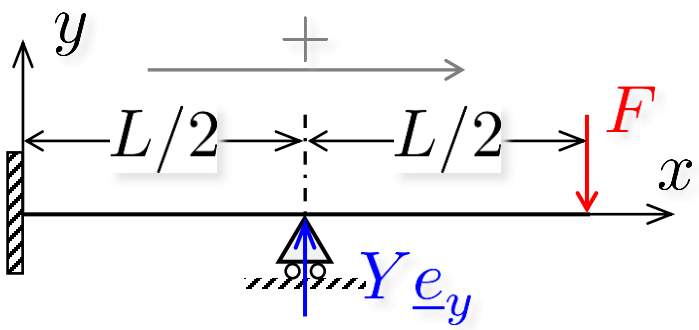
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Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + CL :

$$x \leq L/2 : \theta(x) = -F(Lx - x^2/2)/(EI_z) + Y(Lx/2 - x^2/2)/(EI_z) + \text{cste.} = 0 \; (\theta(0) = 0)$$

$$\xi_n(x) = (-F(Lx^2/2 - x^3/6) + Y(Lx^2/4 - x^3/6))/(EI_z) + \text{cste.}$$

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + Y(L/2-x)$$

Comportement :

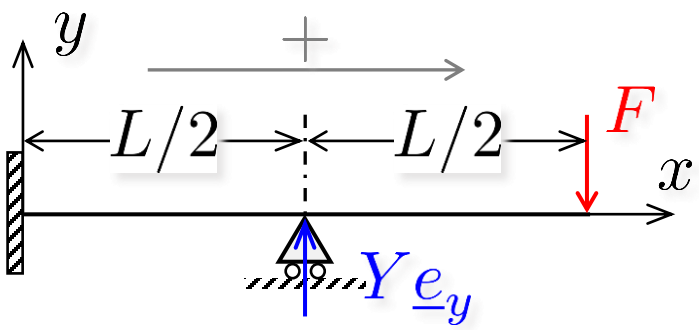
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Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + CL :

$$x \leq L/2 : \theta(x) = -F(Lx - x^2/2)/(EI_z) + Y(Lx/2 - x^2/2)/(EI_z) + \text{cste.} \quad = 0\,(\theta(0) = 0)$$

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Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

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Comportement :

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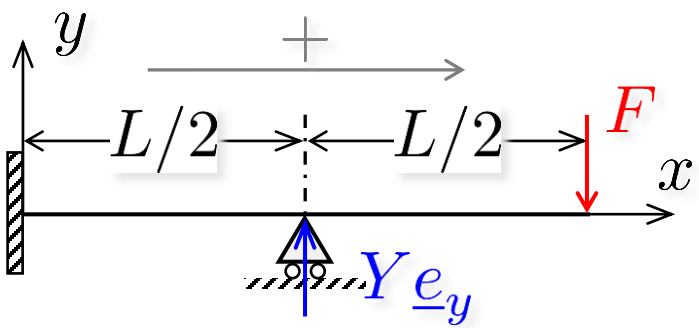
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$$\xi_n(x) = (-F(Lx^2/2 - x^3/6) + Y(Lx^2/4 - x^3/6))/(EI_z) + \text{cste.} \quad = 0 \, (\xi_n(0) = 0)$$

$$\xi_n(L/2) = 0$$

Exemple



Statique :

$$x \geq L/2 : \quad M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + Y(L/2-x)$$

Comportement :

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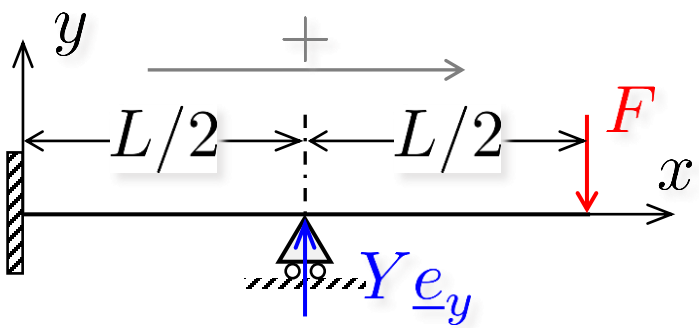
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$$\xi_n(x) = (-F(Lx^2/2 - x^3/6) + Y(Lx^2/4 - x^3/6))/(EI_z) + \text{cste.} \quad = 0 \, (\xi_n(0) = 0)$$

$$\xi_n(L/2) = 0 \longrightarrow \boxed{Y = 5F/2}$$

Exemple (suite) :



$$Y = 5F/2$$

Statique :

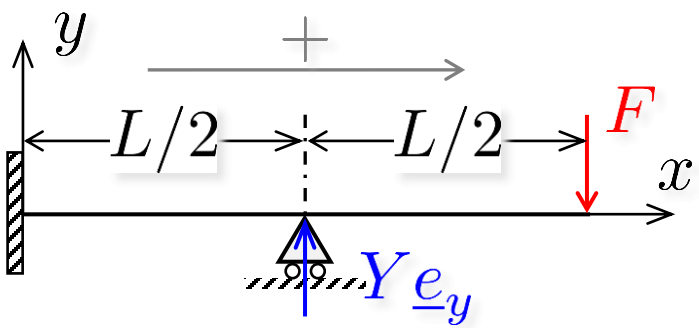
$$x \geq L/2 : \; M = -F(L-x)$$

$$x \leq L/2 : M = -F(L-x) + 5F/2(L/2-x)$$

Comportement :

$$x \geq L/2 : \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x)$$

Exemple (suite) :



$$Y = 5F/2$$

Statique :

$$x \geq L/2 : \; M = -F(L-x)$$

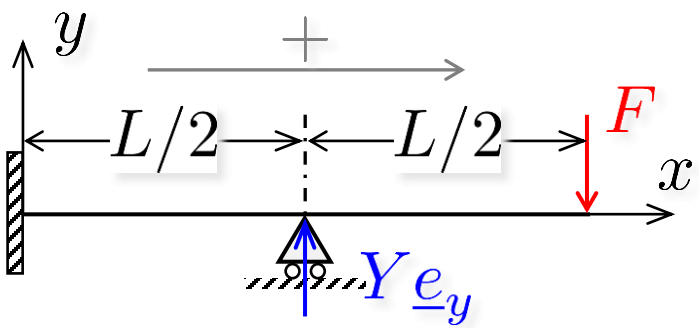
$$x \leq L/2 : M = -F(L-x) + 5F/2(L/2-x)$$

Comportement :

$$x \geq L/2 : \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x)$$

Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + continuité :

Exemple (suite) :



$$Y = 5F/2$$

Statique :

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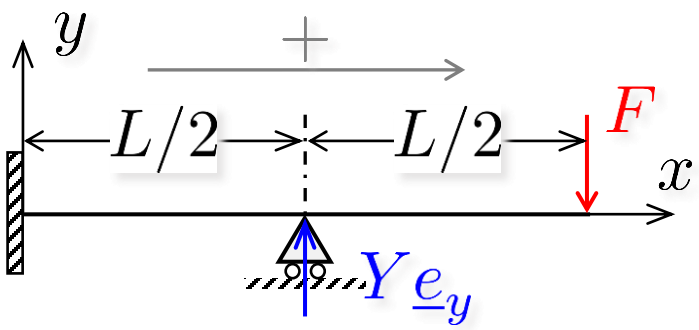
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Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + continuité :

$$x \geq L/2 : \; \theta(x) = -F(Lx - x^2/2)/(EI_z) + \text{cste.}$$

Exemple (suite) :



$$Y = 5F/2$$

Statique :

$$x \geq L/2 : \; M = -F(L-x)$$

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Comportement :

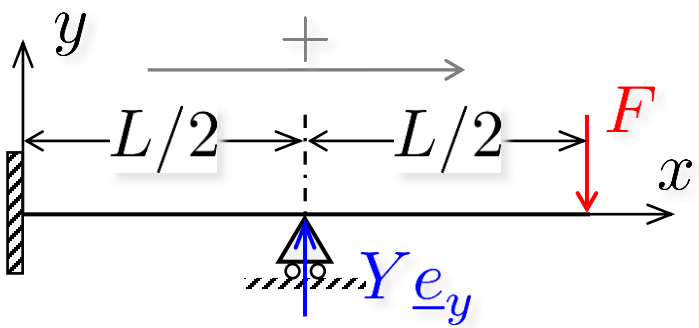
$$x \geq L/2 : \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x)$$

Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + continuité :

$$x \geq L/2 : \theta(x) = -F(Lx - x^2/2)/(EI_z) + \text{cste.}$$

$$= +5FL^2/(16EI_z) \boxed{(\theta(L/2^-) = \theta(L/2^+))}$$

Exemple (suite) :



$$Y = 5F/2$$

Statique :

$$x \geq L/2 : \; M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + 5F/2(L/2-x)$$

Comportement :

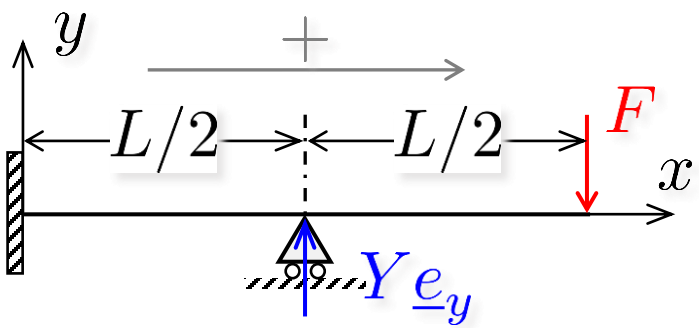
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Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + continuité :

$$x \geq L/2 : \; \theta(x) = -F(Lx - x^2/2)/(EI_z) + \mathrm{cste.} = +5FL^2/(16EI_z) \; \boxed{(\theta(L/2^-) = \theta(L/2^+))}$$

$$\xi_n(x) = -F(Lx^2/2 - x^3/6)/(EI_z) + 5FL^2x/(16EI_z) + \mathrm{cste.}$$

Exemple (suite) :



$$Y = 5F/2$$

Statique :

$$x \geq L/2 : \; M = -F(L-x)$$

$$x \leq L/2 : \; M = -F(L-x) + 5F/2(L/2-x)$$

Comportement :

$$x \geq L/2 : \chi = \frac{M}{EI_z} = \frac{-F}{EI_z}(L-x)$$

Intégration $\left(\chi = \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\mathrm{d}^2\xi_n}{\mathrm{d}x^2}\right)$ + continuité :

$$x \geq L/2 : \; \theta(x) = -F(Lx - x^2/2)/(EI_z) + \text{cste.}$$

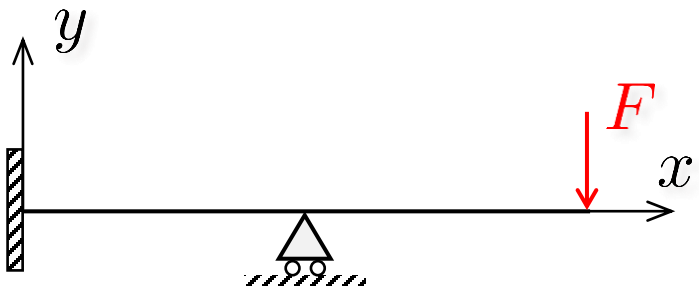
$$= +5FL^2/(16EI_z) \quad (\theta(L/2^-) = \theta(L/2^+))$$

$$\xi_n(x) = -F(Lx^2/2 - x^3/6)/(EI_z) + 5FL^2x/(16EI_z) + \text{cste.}$$

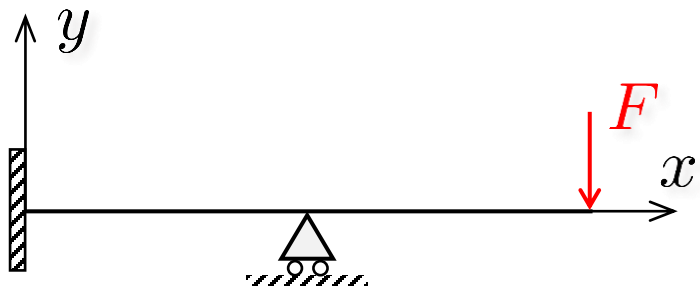
$$= -5FL^3/(96EI_z)$$

$$\xi_n(L/2^-) = \xi_n(L/2^+) = 0$$

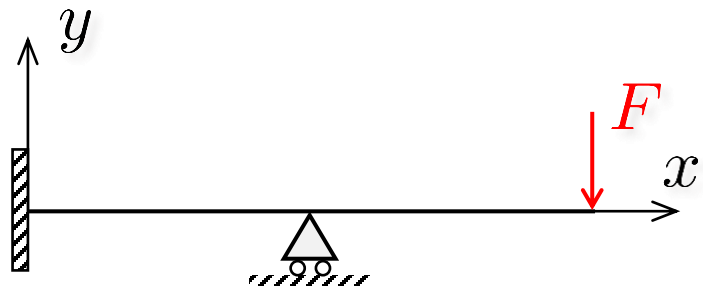
Exemple (suite) :



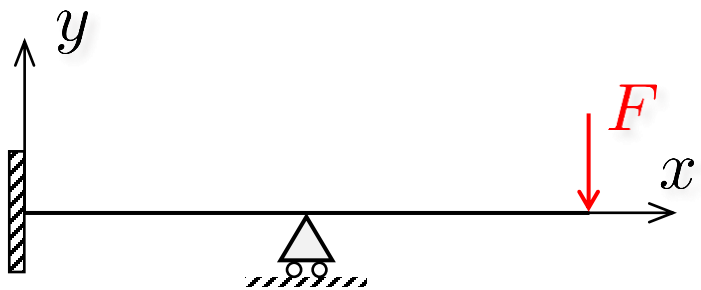
N (effort normal)



V (effort tranchant)

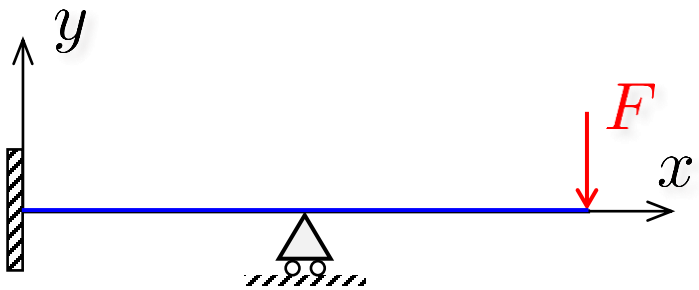


M (moment fléchissant)

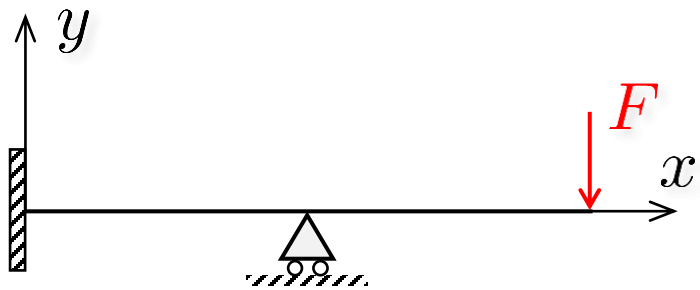


allure de la déformée

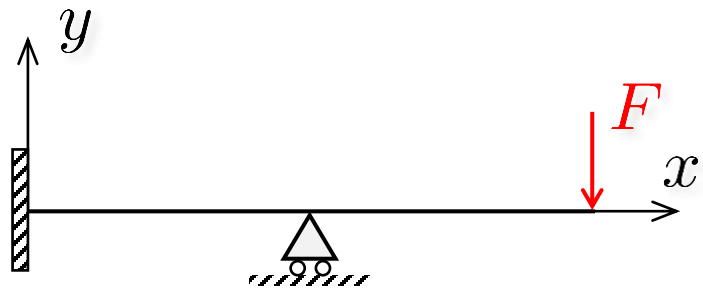
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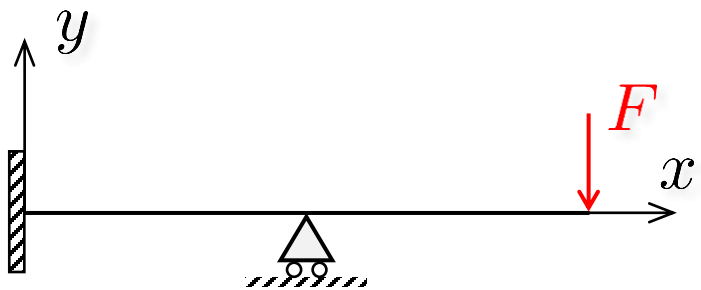
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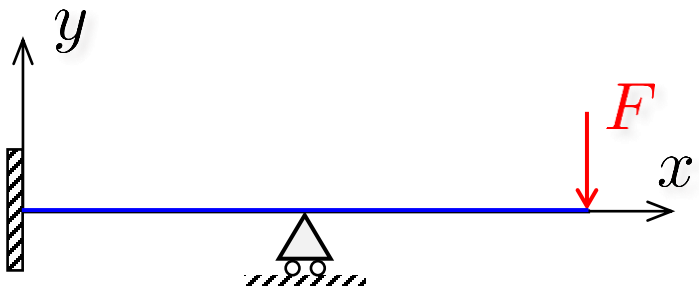


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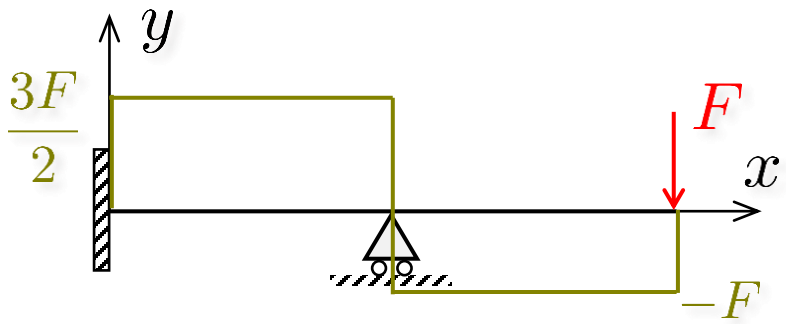


allure de la déformée

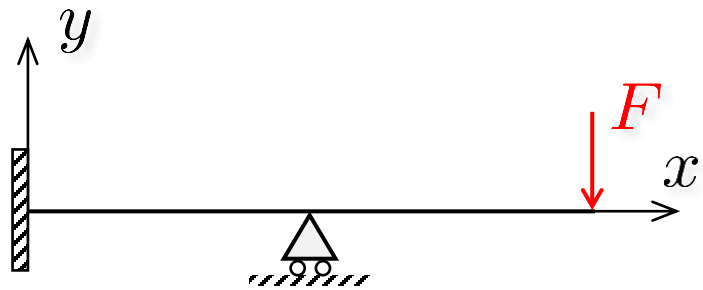
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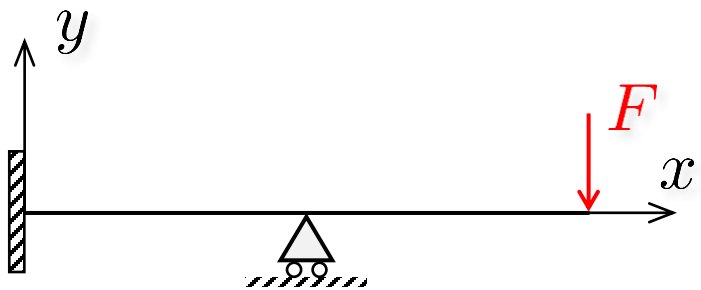
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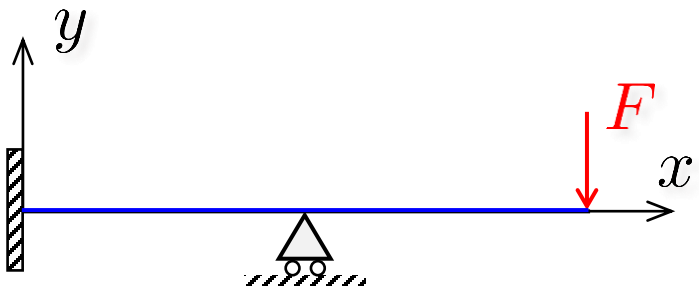


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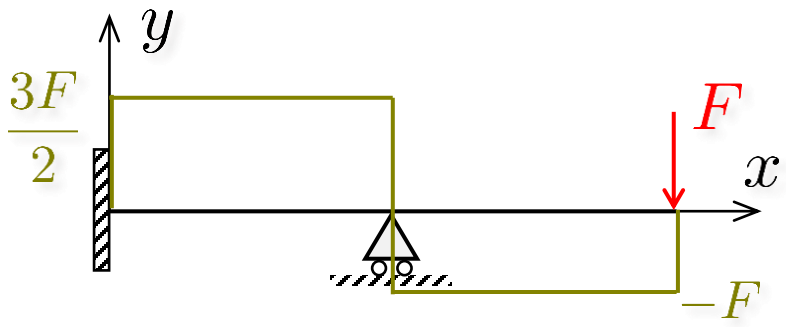


allure de la déformée

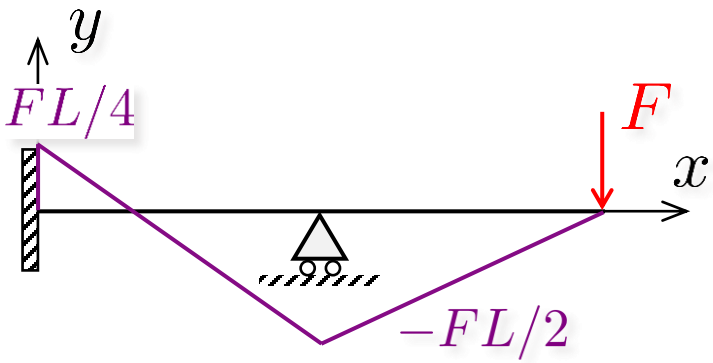
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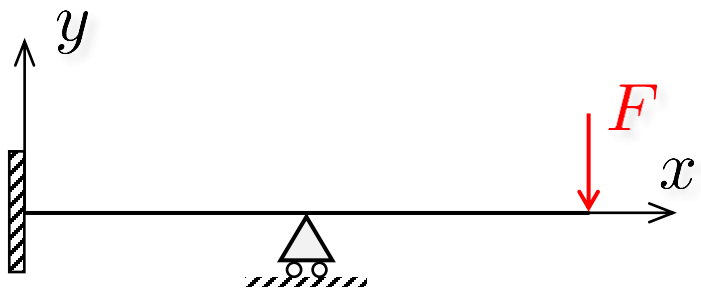
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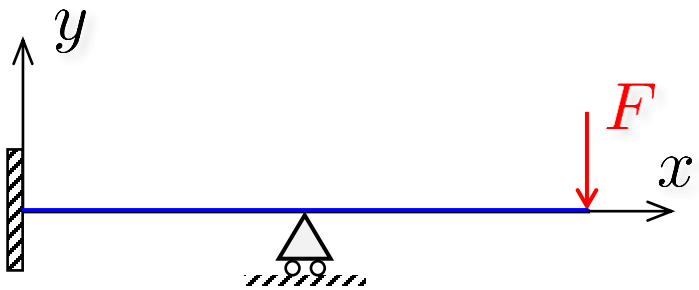


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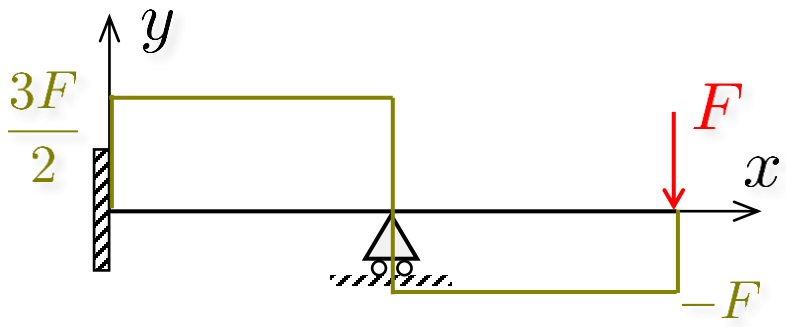


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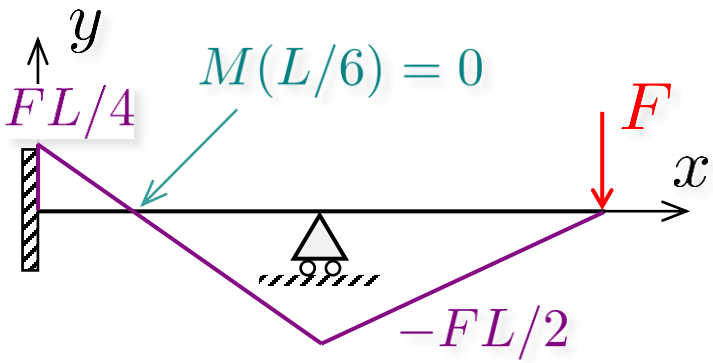
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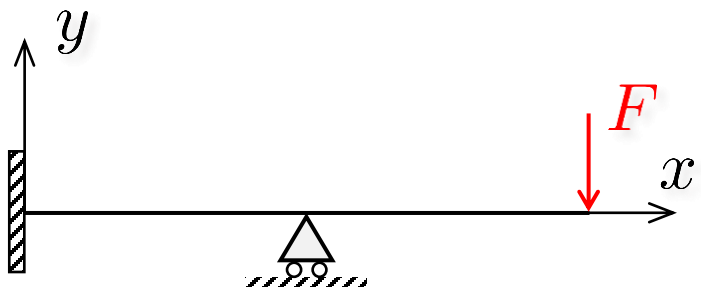
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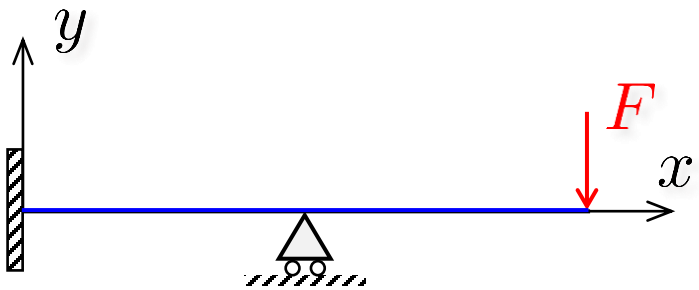


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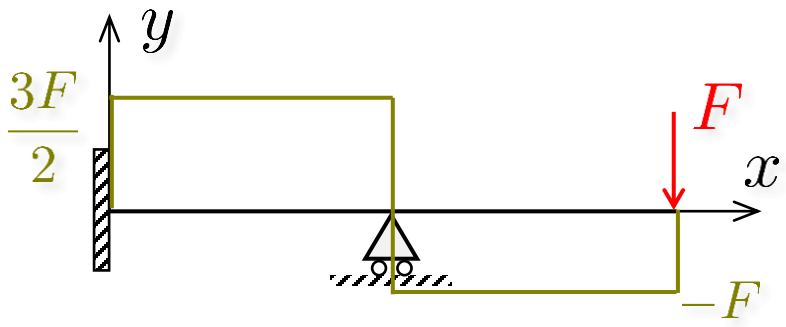


allure de la déformée

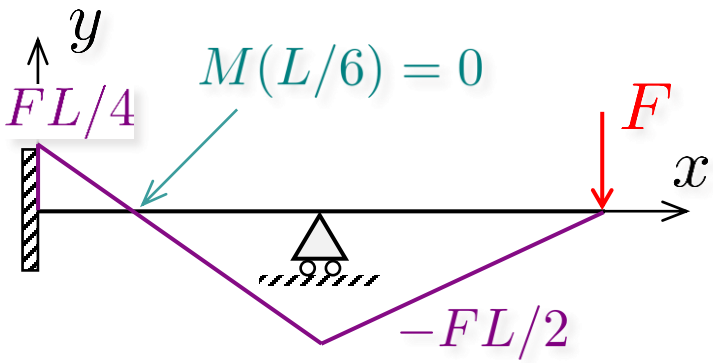
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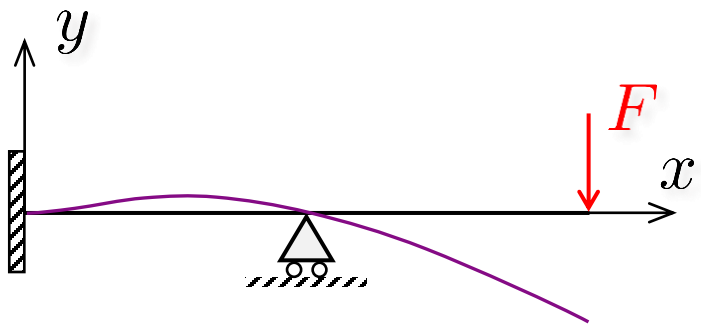
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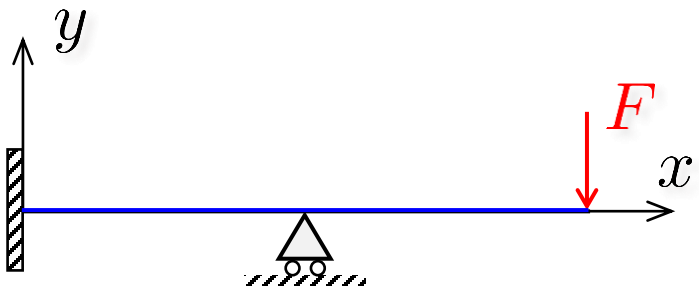


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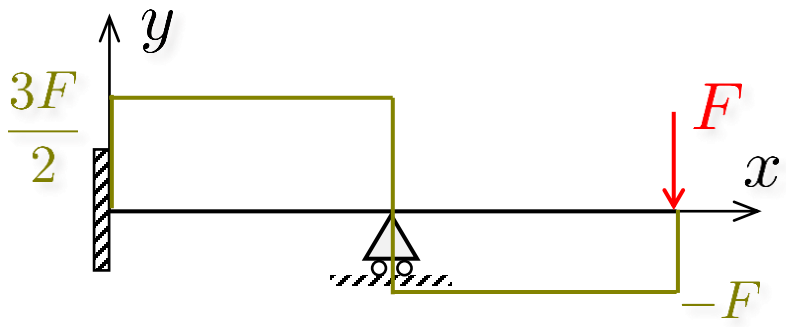


allure de la déformée

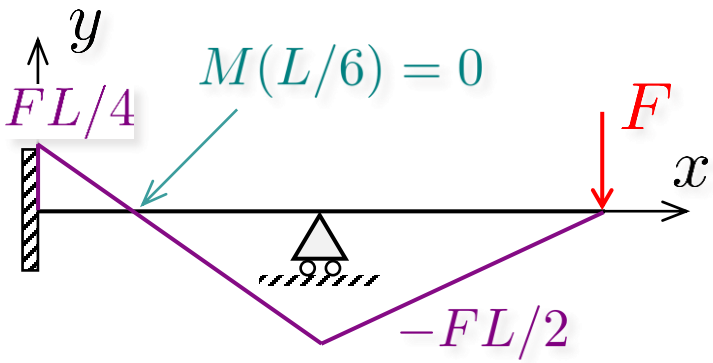
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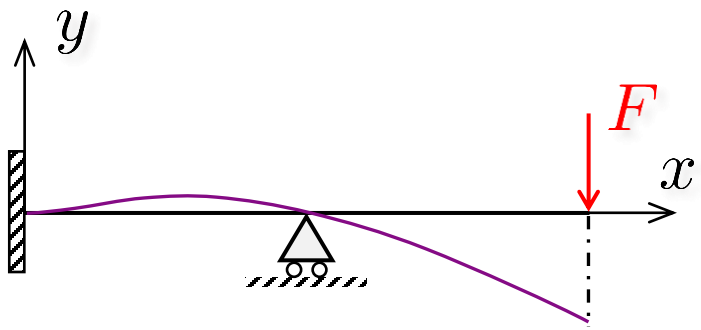
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allure de la déformée